THE LOG-SINE MÖBIUS ROUTE TO THE RIEMANN HYPOTHESIS: WHAT IS PROVED, WHAT IS MEASURED, AND THE ONE INEQUALITY THAT REMAINS

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ABSTRACT. Let $F(u) = \log|2\sin\pi u|$ and $H(x) = \sum_{d\le x}\mu(d)F(x/d)$. We record, with the exact logical status of every statement, the route from H to the Riemann hypothesis followed so far. Proved here: the Mellin transform of F on $[1,\infty)$ is $(\pi/s)\cot(\pi s/2)\zeta(s) + A(s)$ with an explicit A, and A has no zeros in $0 < \operatorname{Re} s < 1$; consequently the Riemann hypothesis is equivalent to $\int_1^X H(x)^2\,dx \ll X^{2+\varepsilon}$, and to $W(X) \ll X^{3/2+\varepsilon}$ for the Clausen sum $W(X) = \sum_{d\le X}\mu(d)\,d\,\mathrm{Cl}_2(2\pi X/d) = -2\pi\int_1^X H$, and also to each of the one-sided bounds $W(X) \le K_\varepsilon X^{3/2+\varepsilon}$ and $W(X) \ge -K_\varepsilon X^{3/2+\varepsilon}$. Unconditionally $W(X)/X^{3/2}$ has $\limsup \ge 0.0821$ and $\liminf \le -0.0821$, the constant being computed numerically from the first zero of ζ. The identity $\mu = 2\mu_U - \mu_U * \mu_U * 1$ on $[1, U^2]$ with $U = \sqrt{X}$ splits W exactly into a part of size $O(X)$, a part $-aX^2 m(\sqrt{X})^2 \le 0$ with $a = \pi(C - \log 2) > 0$, a part $O(X^{3/2+\varepsilon})$, and a corner sum $\mathcal{C}(X)$ that depends only on μ restricted to $[1, \sqrt{X}]$. Hence the Riemann hypothesis is equivalent to the single one-sided inequality $\mathcal{C}(X) \ge -K_\varepsilon X^{3/2+\varepsilon}$. That inequality is not proved. The Müntz transform of the kernel $\mathfrak{k}(t) = F(1/t)\mathbf{1}_{(0,1]}(t)$ is $R(1/t)$, with $R = -E_G'/(2\pi)$, its Möbius dilation sum is $H(1/t)$, and the energy $\int|\sum_{d\le N}\mu(d)\mathfrak{k}(dt)|^2 t^{2\sigma-1}dt$ is an explicit positive semidefinite quadratic form in μ whose uniform boundedness for every $\sigma > \frac{1}{2}$ implies RH (converse not proved). With $\varrho_n = (P\mathfrak{k})(n\,\cdot) - (P\mathfrak{k})/n$ and $\mathcal{H}_\sigma = L^2(t^{2\sigma-1}dt)$, RH is equivalent to $\mathfrak{k}$ lying in the $\mathcal{H}_\sigma$-closure of $\operatorname{span}\{\varrho_n\}$ for every $\sigma \in (\frac{1}{2}, 1)$, a weighted strong Nyman–Beurling criterion for a kernel that is not continuously differentiable. We also record what was measured numerically and which methods were tried and why they do not reach the corner inequality.

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1. Introduction

This document records the state of one line of work. Its purpose is to say exactly what has been proved, in which direction, and under which hypothesis, and to separate it from what has only been measured. Nothing in it is used in a direction that has not been proved. The Riemann hypothesis (RH) is used in the form: $\zeta(s) \neq 0$ for $\operatorname{Re} s > \frac{1}{2}$.

Throughout, L^1 and L^2 are the Lebesgue spaces, $F(u) = \log|2\sin \pi u|$ for $u \in \mathbb{R} \setminus \mathbb{Z}$, μ is the Möbius function, $M(x) = \sum_{n \le x} \mu(n)$, $m(x) = \sum_{n \le x} \mu(n)/n$, and Cl_2 is the Clausen function (2.1). The constant

$$C = -\int_1^\infty \log|\sin \pi u|\, u^{-2}\, du = 0.74379486183940054\ldots \tag{1.1}$$

and $a = \pi(C - \log 2) = 0.15911438342889\ldots$ (Corollary 2.3).

The table below lists every item and its status. "Proved" means proved in this document, with the classical inputs cited at the step where they are used. "Measured" means a floating-point computation, reported with its range, with no claim beyond it. "Not proved" means exactly that.

Item	Status
Mellin transform of F, Theorem 2.2	proved
$A(s) \neq 0$ for $0 < \operatorname{Re} s < 1$, Theorem 3.1	proved
Local structure of H and the two identities, Propositions 4.1, 4.2	proved
RH $\iff \int_1^X H^2 \ll X^{2+\varepsilon}$, Theorem 4.4	proved, both directions
RH $\iff W \ll X^{3/2+\varepsilon} \iff$ either one-sided bound, Theorem 5.3	proved, all directions
$\Omega_\pm$ bounds for $W/X^{3/2}$, Theorem 5.4	proved; the constant 0.0821 is computed numerically
Exact $d = 2$ decomposition of W, Proposition 6.3	proved
$E_G(Y) \ll_\delta Y^{1+\delta}$, Lemma 6.2	proved
RH $\iff \mathcal{C}(X) \geq -K_\varepsilon X^{3/2+\varepsilon} \iff \lvert\mathcal{C}(X)\rvert \ll X^{3/2+\varepsilon}$, Theorem 6.4	proved, all directions
Other exact identities (lcm form, recursion over large primes, $d = 3$), Section 7	proved
$W_{\mu^2}(X) = (6/\pi^2)aX^2 + O(X^{3/2+\delta})$, Proposition 8.1	proved
Müntz form of R and H, energy and Gram matrix, Propositions 9.1, 9.2	proved
$\liminf_N \mathcal{E}_N(\sigma) < \infty\ \forall \sigma > \frac{1}{2} \Rightarrow$ RH, Theorem 9.3	proved, one direction only
Weighted strong Nyman–Beurling criterion, Theorem 10.3	proved, both directions
Finite prime cut $\Rightarrow$ RH, Proposition 10.4	proved, one direction; converse **not proved**
Liouville form of the corner and Clausen–Müntz structure, Propositions 11.1, 11.2	proved
One-sided Liouville–Clausen criterion, Proposition 11.3	proved, one direction
$\mathcal{C}_{\mu^2} \sim \kappa X^2$, Remark 11.5	constant computed; asymptotic **not proved**, not confirmed by the data
Corner inequality (14.1)	**not proved**
Off-diagonal bound (9.12)	**not proved**; sufficient for RH, not known to be necessary
Localization inequality of Remark 8.3	**not proved**, not used
Tables of Section 13	measured

Every criterion below is a reformulation of RH, and it is stated with both directions proved. The reduction of Section 6 replaces the two-sided bound for W by a one-sided bound for a sum that involves μ only on $[1, \sqrt{X}]$. The direction used to reach RH, corner bound $\Rightarrow$ RH, needs no

information on $m(\sqrt{X})$ or on any tail of m: the term $-aX^2 m(\sqrt{X})^2$ has a sign and is simply dropped. Information on m is used only in the converse direction, RH $\Rightarrow$ corner bound.

2. The Mellin transform of the log-sine function

For $\theta \in \mathbb{R}$ let

$$\mathrm{Cl}_2(\theta) = -\int_0^\theta \log\Big|2\sin\frac{\psi}{2}\Big|\,d\psi = \sum_{\nu\ge 1}\frac{\sin\nu\theta}{\nu^2}. \tag{2.1}$$

The second equality is (2.11) below with $\theta = 2\pi u$. The function Cl_2 is odd, 2π-periodic and continuous. Its derivative $-\log|2\sin(\theta/2)|$ vanishes on $(0,\pi)$ only at $\theta = \pi/3$, is positive on $(0,\pi/3)$ and negative on $(\pi/3, 5\pi/3)$. Hence

$$|\mathrm{Cl}_2(\theta)| \le \mathrm{Cl}_2(\pi/3) = 1.01494160640965\ldots \qquad (\theta\in\mathbb{R}). \tag{2.2}$$

Reviewed by Claude (review 1)

Lemma 2.1 (Euler integrals)**.** *Let $\alpha > 0$. For $-1 < \operatorname{Re} w < 0$ the integral below converges absolutely and*

$$\int_0^\infty u^{w-1}\sin(\alpha u)\,du = \alpha^{-w}\,\Gamma(w)\sin\frac{\pi w}{2}. \tag{2.3}$$

Proof. First let $0 < \operatorname{Re} z < 1$. We show that

$$\lim_{R\to\infty}\int_0^R u^{z-1}e^{\pm iu}\,du = \Gamma(z)\,e^{\pm i\pi z/2}. \tag{2.4}$$

Take the upper sign. Integrate $\xi^{z-1}e^{i\xi}$, with the principal branch, over the boundary of $\{\eta \le |\xi| \le R,\ 0 \le \arg\xi \le \pi/2\}$, where $0 < \eta < R$. Cauchy's theorem gives 0. On the arc $\xi = Re^{i\theta}$ we have $|e^{i\xi}| = e^{-R\sin\theta} \le e^{-2R\theta/\pi}$ and $|\xi^{z-1}| \le R^{\operatorname{Re} z-1}e^{\pi|\operatorname{Im} z|/2}$, so the arc contributes at most

$$e^{\pi|\operatorname{Im} z|/2}R^{\operatorname{Re} z}\int_0^{\pi/2} e^{-2R\theta/\pi}\,d\theta \le e^{\pi|\operatorname{Im} z|/2}\,\frac{\pi}{2}\,R^{\operatorname{Re} z-1}, \tag{2.5}$$

which tends to 0 because $\operatorname{Re} z < 1$. The arc of radius η contributes at most $e^{\pi|\operatorname{Im} z|/2}\frac{\pi}{2}\eta^{\operatorname{Re} z}$, which tends to 0 as $\eta\to 0$ because $\operatorname{Re} z > 0$. On the imaginary axis $\xi = iy$ the integral is $i^z\int_0^\infty y^{z-1}e^{-y}\,dy = e^{i\pi z/2}\Gamma(z)$. This proves (2.4) with the upper sign. For the lower sign integrate $\xi^{z-1}e^{-i\xi}$ over the boundary of $\{\eta \le |\xi| \le R,\ -\pi/2 \le \arg\xi \le 0\}$. On the arc $\xi = Re^{i\theta}$, $-\pi/2 \le \theta \le 0$, we have $|e^{-i\xi}| = e^{R\sin\theta} \le e^{-2R|\theta|/\pi}$, so both arcs obey the bounds above, and on the axis $\xi = -iy$ the integral is $(-i)^z\Gamma(z) = e^{-i\pi z/2}\Gamma(z)$. Half the sum and half the difference divided by i give $\int_0^\infty u^{z-1}\cos u\,du = \Gamma(z)\cos(\pi z/2)$ and $\int_0^\infty u^{z-1}\sin u\,du = \Gamma(z)\sin(\pi z/2)$ as improper integrals.

Now let $-1 < \operatorname{Re} w < 0$. The integrand in (2.3) with $\alpha = 1$ is $O(u^{\operatorname{Re} w})$ near 0 and $O(u^{\operatorname{Re} w-1})$ at infinity, so it is absolutely integrable. Integration by parts gives

$$\int_0^\infty u^{w-1}\sin u\,du = \Big[\frac{u^w\sin u}{w}\Big]_0^\infty - \frac{1}{w}\int_0^\infty u^w\cos u\,du = -\frac{1}{w}\,\Gamma(w+1)\cos\frac{\pi(w+1)}{2}, \tag{2.6}$$

where $w \ne 0$ because $\operatorname{Re} w < 0$, the boundary terms vanish because $-1 < \operatorname{Re} w < 0$, and the last integral is the cosine case with $z = w+1$, $0 < \operatorname{Re} z < 1$. Since $\Gamma(w+1) = w\Gamma(w)$ and $\cos(\pi(w+1)/2) = -\sin(\pi w/2)$, this is $\Gamma(w)\sin(\pi w/2)$. The substitution $u \mapsto \alpha u$ gives the factor α^{-w}. □

For $\operatorname{Re} s < 2$, $s \ne 0$, let

$$A(s) = \frac{\log(2\pi)}{s} + \frac{1}{s^2} + \sum_{k\ge 1}\frac{\zeta(2k)}{k\,(2k-s)}. \tag{2.7}$$

The series converges absolutely and locally uniformly on $\operatorname{Re} s < 2$, because $\zeta(2k) \le \zeta(2)$ and $|2k-s| \ge 2k - \operatorname{Re} s \ge 2(k-1) + (2-\operatorname{Re} s)$; so A is holomorphic on $\{\operatorname{Re} s < 2\}\setminus\{0\}$.

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Theorem 2.2 (Mellin transform of F)**.** *For* $\operatorname{Re} s > 0$ *the integral*

$$\widehat{F}(s) = \int_1^\infty F(u)\, u^{-s-1}\, du \tag{2.8}$$

converges absolutely. The function $\widehat{F}$ *extends holomorphically to* $\operatorname{Re} s > -1$*, and for* $0 < \operatorname{Re} s < 2$

$$\widehat{F}(s) = \frac{\pi}{s}\cot\Big(\frac{\pi s}{2}\Big)\zeta(s) + A(s), \tag{2.9}$$

where at $s = 1$ *the first term is defined by continuity, with value* $-\pi^2/2$*.*

Proof. Absolute convergence for $\operatorname{Re} s > 0$ holds because F is 1-periodic and integrable over a period. For $0 < \lambda < 1$ and real v one has $\log|1 - \lambda e^{2\pi i v}| = -\sum_{\nu\geq 1} \lambda^\nu \cos(2\pi\nu v)/\nu$. As $\lambda \to 1^-$ the left side tends to $F(v)$ in $L^2(0,1)$ by dominated convergence, and the Fourier coefficients on the right tend to $-1/(2\nu)$. Hence

$$F(v) = -\sum_{\nu\geq 1} \frac{\cos 2\pi\nu v}{\nu} \qquad \text{in } L^2(0,1). \tag{2.10}$$

In particular $\int_0^1 F = 0$ and, by Parseval, $\int_0^1 F^2 = \sum_\nu 1/(2\nu^2) = \pi^2/12$. Integrating (2.10) over $[0,u]$ and using periodicity, the function $\Phi(u) = \int_0^u F(v)\, dv$ satisfies

$$\Phi(u) = -\frac{1}{2\pi}\sum_{\nu\geq 1} \frac{\sin 2\pi\nu u}{\nu^2} = -\frac{\operatorname{Cl}_2(2\pi u)}{2\pi} \qquad (u \geq 0), \tag{2.11}$$

with an absolutely and uniformly convergent series. The second equality is the definition (2.1) after the substitution $\psi = 2\pi v$. Thus Φ is bounded, $\Phi(n) = 0$ for every integer $n \geq 0$, and, since $2\sin\pi v \geq 4v$ on $[0, \frac{1}{2}]$,

$$|\Phi(u)| \leq \int_0^u \log\frac{1}{4v}\, dv = u\Big(\log\frac{1}{4u} + 1\Big) \qquad (0 < u \leq \tfrac{1}{6}). \tag{2.12}$$

By parts, using $\Phi(1) = 0$ and the boundedness of Φ, for $\operatorname{Re} s > 0$

$$\widehat{F}(s) = (s+1)\int_1^\infty \Phi(u)\, u^{-s-2}\, du, \tag{2.13}$$

and the right-hand side converges absolutely and is holomorphic for $\operatorname{Re} s > -1$. This gives the extension.

Let $\sigma = \operatorname{Re} s \in (-1, 0)$. Integration by parts, with boundary terms vanishing by (2.12) at 0 and by boundedness of Φ at infinity, gives the convergent improper integral

$$\mathcal{L}(s) := \int_0^\infty F(u)\, u^{-s-1}\, du = (s+1)\int_0^\infty \Phi(u)\, u^{-s-2}\, du. \tag{2.14}$$

Substituting (2.11) and integrating termwise is justified by Fubini's theorem, because the series of absolute values converges by Tonelli's theorem:

$$\sum_{\nu\geq 1} \frac{1}{\nu^2}\int_0^\infty |\sin 2\pi\nu u|\, u^{-\sigma-2}\, du = b_\sigma (2\pi)^{\sigma+1} \sum_{\nu\geq 1} \nu^{\sigma-1} < \infty, \tag{2.15}$$

where $b_\sigma = \int_0^\infty |\sin v|\, v^{-\sigma-2}\, dv$ is finite because $-1 < \sigma < 0$. Lemma 2.1 with $w = -s-1$ and $\alpha = 2\pi\nu$, together with $(s+1)\Gamma(-s-1) = -\Gamma(-s)$ and $\sin(-\pi(s+1)/2) = -\cos(\pi s/2)$, gives

$$\mathcal{L}(s) = -(2\pi)^s\, \Gamma(-s)\cos\Big(\frac{\pi s}{2}\Big)\zeta(1-s) = \frac{\pi}{s}\cot\Big(\frac{\pi s}{2}\Big)\zeta(s) \qquad (-1 < \sigma < 0). \tag{2.16}$$

The second equality uses the functional equation $\zeta(1-s) = 2(2\pi)^{-s}\cos(\pi s/2)\Gamma(s)\zeta(s)$ [5, (2.1.8), p. 16] and Euler's reflection formula, which gives $\Gamma(s)\Gamma(-s) = -\pi/(s\sin\pi s)$.

Split $\mathcal{L}(s) = \widehat{F}(s) + \mathcal{I}(s)$ according to $u \geq 1$ and $u < 1$; here $\widehat{F}$ is given by (2.13), and the split is legitimate because (2.14) and (2.13) are obtained by the same integration by parts on $[1, \infty)$. For $\operatorname{Re} s < 0$, writing $F(u) = \log(2\pi u) + \log(\sin \pi u/(\pi u))$ on $(0, 1)$,

$$\mathcal{I}(s) := \int_0^1 F(u)\, u^{-s-1}\, du = -\frac{\log(2\pi)}{s} - \frac{1}{s^2} + \mathcal{J}(s), \tag{2.17}$$

$$\mathcal{J}(s) := \int_0^1 \log\Big(\frac{\sin \pi u}{\pi u}\Big) u^{-s-1}\, du = -\sum_{k \geq 1} \frac{\zeta(2k)}{k\,(2k-s)} \qquad (\operatorname{Re} s < 2). \tag{2.18}$$

For (2.18) we use Euler's product for the sine, which gives, for $0 \leq u < 1$,

$$\log \frac{\sin \pi u}{\pi u} = \sum_{\nu \geq 1} \log\Big(1 - \frac{u^2}{\nu^2}\Big) = -\sum_{k \geq 1} \frac{\zeta(2k)}{k}\, u^{2k}. \tag{2.19}$$

All terms have the same sign, and $\sum_k \zeta(2k)/(k(2k - \operatorname{Re} s)) < \infty$ for $\operatorname{Re} s < 2$, so termwise integration is justified by Fubini's theorem, the absolute series being bounded through Tonelli's theorem, and $\mathcal{J}$ is holomorphic for $\operatorname{Re} s < 2$.

Combining (2.16) and (2.17), for $-1 < \operatorname{Re} s < 0$,

$$\widehat{F}(s) = \frac{\pi}{s} \cot\Big(\frac{\pi s}{2}\Big)\zeta(s) + \frac{\log(2\pi)}{s} + \frac{1}{s^2} - \mathcal{J}(s). \tag{2.20}$$

By (2.7) and (2.18), the last three terms are $A(s)$. Both sides are meromorphic on the connected strip $-1 < \operatorname{Re} s < 2$, so (2.20) holds there by the identity theorem, and this is (2.9). At $s = 1$, $\cot(\pi s/2) = -\tan(\pi(s-1)/2)$ and $(s-1)\zeta(s) \to 1$ [5, (2.1.16), p. 16], so the first term tends to $-\pi^2/2$. □

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Corollary 2.3 (Two forms of $sA(s)$, and the constants)**.** *Let $B(s) = sA(s)$. For $\operatorname{Re} s < 2$, $s \neq 0$,*

$$B(s) = \log(2\pi) + \frac{1}{s} + \sum_{k \geq 1} \zeta(2k)\Big(\frac{2}{2k-s} - \frac{1}{k}\Big), \tag{2.21}$$

$$B(s) = \log \pi + \frac{1}{s} + \sum_{k \geq 1} \frac{s/k + 2(\zeta(2k) - 1)}{2k - s}, \tag{2.22}$$

with absolutely convergent series. Moreover $\widehat{F}(1) = \log 2 - C$, and

$$a := \pi(C - \log 2) = -\pi \widehat{F}(1) = 0.15911438342889\ldots > 0. \tag{2.23}$$

Proof. For $k \geq 1$, $s/(k(2k-s)) = 2/(2k-s) - 1/k$; multiplying (2.7) by s gives (2.21), whose terms are $\zeta(2k)s/(k(2k-s)) = O(k^{-2})$. Next, $\sum_{k \geq 1}(\zeta(2k) - 1)/k = \sum_{\nu \geq 2} -\log(1 - \nu^{-2})$ by Tonelli's theorem, and $\prod_{\nu=2}^{N} \nu^2/((\nu-1)(\nu+1)) = 2N/(N+1) \to 2$ as $N \to \infty$, so

$$\sum_{k \geq 1} \frac{\zeta(2k) - 1}{k} = \log 2. \tag{2.24}$$

Writing $\zeta(2k) = 1 + (\zeta(2k) - 1)$ in (2.21), the part with 1 is $\sum_k (2/(2k-s) - 1/k) = \sum_k (s/k)/(2k-s)$, and the part with $\zeta(2k) - 1$ is $\sum_k 2(\zeta(2k) - 1)/(2k-s) - \log 2$ by (2.24). This gives (2.22). Finally, $-\log|\sin \pi u| = \log 2 - F(u)$ and $\int_1^\infty u^{-2}\, du = 1$, so (1.1) gives $C = \log 2 - \widehat{F}(1)$. The numerical values of C and a are computed from (2.9) at $s = 1$, which gives $\widehat{F}(1) = -\pi^2/2 + A(1)$, by the script `route_constants.py`; the same script checks (2.9) against direct numerical integration of (2.13) at $s = \frac{1}{2} + 3i$ and $s = \frac{3}{2} + 2i$ (agreement to 10^{-8}, the size of the truncation of the integral). □

3. The function A has no zeros in the critical strip

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Theorem 3.1. *$A(s) \neq 0$ for $0 < \operatorname{Re} s < 1$. More precisely, with $s = \sigma + it$, $0 < \sigma < 1$,*

$$\operatorname{Re} B(s) > \log \pi - \frac{\zeta(3)}{2} = 0.54370\ldots \qquad (|t| \le \sqrt{2}), \tag{3.1}$$

$$\frac{\operatorname{Im} B(s)}{t} = -\frac{1}{\sigma^2 + t^2} + \sum_{k \ge 1} \frac{2\zeta(2k)}{(2k-\sigma)^2 + t^2} > 0 \qquad (|t| \ge \sqrt{2}). \tag{3.2}$$

Proof. Since $A = B/s$ and $s \neq 0$, $A(s) = 0$ if and only if $B(s) = 0$. We show $B(s) \neq 0$.

Case $|t| \le \sqrt{2}$. Use (2.22). For $k \ge 1$ the term $2(\zeta(2k) - 1)/(2k - s)$ has real part $2(\zeta(2k) - 1)(2k - \sigma)/|2k - s|^2 \ge 0$, because $\zeta(2k) > 1$ and $2k - \sigma > 0$. Also $\operatorname{Re}(1/s) = \sigma/|s|^2 > 0$. For the term $s/(k(2k - s))$ we use the identity, valid for all real σ, t and $k \ge 1$,

$$\operatorname{Re} \frac{s}{2k - s} + \frac{t^2}{4k^2 + t^2} = \frac{2k\sigma\big(2k(2k - \sigma) - t^2\big)}{\big((2k - \sigma)^2 + t^2\big)\big(4k^2 + t^2\big)}. \tag{3.3}$$

To check it, $\operatorname{Re}(s/(2k - s)) = (2k\sigma - \sigma^2 - t^2)/D_1$ with $D_1 = (2k - \sigma)^2 + t^2$; with $D_2 = 4k^2 + t^2$ the numerator of the left side over $D_1 D_2$ is $(2k\sigma - \sigma^2)D_2 - t^2 D_2 + t^2 D_1 = \sigma(2k - \sigma)(4k^2 + t^2) - \sigma t^2(4k - \sigma) = 2k\sigma(2k(2k - \sigma) - t^2)$, since $D_1 - D_2 = -4k\sigma + \sigma^2$. For $0 < \sigma < 1$, $k \ge 1$ and $t^2 \le 2$ we have $2k(2k - \sigma) > 2k(2k - 1) \ge 2 \ge t^2$, so the right side of (3.3) is positive and

$$\operatorname{Re} \frac{s}{k(2k - s)} > -\frac{t^2}{k(4k^2 + t^2)} \ge -\frac{t^2}{4k^3} \ge -\frac{1}{2k^3}. \tag{3.4}$$

Summing over k gives (3.1), since $\log \pi - \zeta(3)/2 = 1.14473\ldots - 0.60103\ldots = 0.54370\ldots$.

Case $|t| \ge \sqrt{2}$. Use (2.21). The terms $\log(2\pi)$ and $-\zeta(2k)/k$ are real, $\operatorname{Im}(1/s) = -t/(\sigma^2 + t^2)$ and $\operatorname{Im}(2/(2k - s)) = 2t/((2k - \sigma)^2 + t^2)$, which gives the equality in (3.2). All terms of the sum are positive, and the term $k = 1$ alone exceeds $1/(\sigma^2 + t^2)$, because $2\zeta(2) = \pi^2/3$ and

$$\frac{\pi^2}{3}(\sigma^2 + t^2) - (2 - \sigma)^2 - t^2 \ge \Big(\frac{\pi^2}{3} - 1\Big)t^2 - (2 - \sigma)^2 \ge 2\Big(\frac{\pi^2}{3} - 1\Big) - 4 = \frac{2\pi^2}{3} - 6 = 0.5797\ldots > 0 \tag{3.5}$$

for $t^2 \ge 2$ and $0 < \sigma < 1$. Hence $\operatorname{Im} B(s) \neq 0$. □

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Remark 3.2. The identities (3.3) and (3.2), and the expansion behind (3.5), are also checked symbolically by `route_constants.py`. At the first zero $\rho_1 = \frac{1}{2} + i\,14.1347\ldots$ of ζ one finds numerically $A(\rho_1) = 0.10741008 + 0.10085232\,i$, $|A(\rho_1)| = 0.147337$.

4. The Möbius log-sine function H

For $x \ge 1$, $x \notin \mathbb{Z}$, let

$$H(x) = \sum_{d \le x} \mu(d)\, F\Big(\frac{x}{d}\Big). \tag{4.1}$$

Each x/d is then not an integer, so every term is finite.

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Proposition 4.1 (Local structure)**.** (i) *For $X \ge 1$,*

$$\int_1^X |H(x)|^2\, dx \le \frac{\pi^2}{12} X^3, \qquad \int_1^X |H(x)|\, dx \le \frac{\pi}{2\sqrt{3}} X^2. \tag{4.2}$$

(ii) *Let $n \ge 2$ be an integer. As $x \to n^-$, $H(x) + \mu(n)\log(n - x)$ has a finite limit. As $x \to n^+$, $H(x)$ has the finite limit*

$$H(n^+) = \Lambda(n) + \sum_{d \le n,\ d \nmid n} \mu(d)\, F\Big(\frac{n}{d}\Big). \tag{4.3}$$

Proof. (i) For $1 \le d \le X$, by periodicity, $\int_d^X F(x/d)^2\,dx = d\int_1^{X/d} F^2 \le d\lceil X/d-1\rceil \int_0^1 F^2 \le X\pi^2/12$, because $[1, X/d]$ is covered by $\lceil X/d-1\rceil \le X/d$ intervals of length 1. By Cauchy–Schwarz over the at most $\lfloor x\rfloor \le X$ terms, $|H(x)|^2 \le X\sum_{d\le x} F(x/d)^2$. Integrating over $[1, X]$ and summing the bound above over the at most X values of d gives $\int_1^X |H|^2 \le X\cdot X\cdot X\pi^2/12$, the first bound. For the second, $\int_d^X |F(x/d)|\,dx \le d\lceil X/d-1\rceil\int_0^1 |F| \le X(\pi^2/12)^{1/2}$, and summing over $d \le X$ gives the second bound, since $(\pi^2/12)^{1/2} = \pi/(2\sqrt{3})$.

(ii) Let $x \in (n-1, n)$. The terms of (4.1) are those with $d \le n-1$. A term with $d \nmid n$ is continuous at $x = n$. For $d \mid n$, $d < n$, we have $F(x/d) = \log(2\pi(n-x)/d) + o(1)$ as $x \to n^-$, because $|\sin(\pi x/d)| = |\sin(\pi(n-x)/d)|$. Since $\sum_{d|n,\, d<n}\mu(d) = -\mu(n)$ for $n \ge 2$, the singular part is $-\mu(n)\log(n-x)$. For $x \in (n, n+1)$ the term $d = n$ is present too, the singular parts of the $d \mid n$ terms add up to $\sum_{d|n}\mu(d)\log(2\pi(x-n)) = 0$, and the remaining constants are $-\sum_{d|n}\mu(d)\log d = \Lambda(n)$. □

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Proposition 4.2 (Two identities)**.** *Let $x > 1$, $x \notin \mathbb{Z}$, and let $\log(1-e^{2\pi i v})$ denote the principal branch. Then*

$$\sum_{d\le x}\mu(d)\log(1-e^{2\pi i x/d}) = H(x) + i\pi\Big(x\,m(x) - 1 - \frac{M(x)}{2}\Big). \tag{4.4}$$

With $\operatorname{sinc} v = \sin(\pi v)/(\pi v)$,

$$H(x) = \log|x^2-1| + \log(2\pi)M(x) + \int_1^x \frac{M(u)}{u}\,du - \sum_{d>x}\mu(d)\log\operatorname{sinc}\frac{x}{d}, \tag{4.5}$$

where the last series converges absolutely.

Proof. For $v > 0$, $v \notin \mathbb{Z}$, $\log(1-e^{2\pi i v}) = F(v) + i\pi(\{v\} - \frac12)$. Summing with weights $\mu(d)$ at $v = x/d$, and using $\{x/d\} = x/d - \lfloor x/d\rfloor$ and $\sum_{d\le x}\mu(d)\lfloor x/d\rfloor = 1$ for $x \ge 1$, gives (4.4).

For (4.5), Euler's product gives $\log|\operatorname{sinc} v| = \sum_{\nu\ge1}\log|1 - v^2/\nu^2|$ for $v \notin \mathbb{Z}$. For fixed x, $|\log|1-x^2/n^2|| \le 2x^2/n^2$ when $n \ge 2x$, and the pairs (d,ν) with $d\nu < 2x$ are finitely many, so $\sum_{d,\nu}|\log|1-x^2/(d\nu)^2|| \le \sum_n \tau(n)|\log|1-x^2/n^2|| < \infty$, where $\tau(n)$ is the number of divisors of n. Grouping by $n = d\nu$ and using $\sum_{d|n}\mu(d) = [n=1]$,

$$\sum_{d\ge1}\mu(d)\log\Big|\operatorname{sinc}\frac{x}{d}\Big| = \log|1-x^2|. \tag{4.6}$$

For $d \le x$, $F(x/d) = \log(2\pi x/d) + \log|\operatorname{sinc}(x/d)|$, and $\sum_{d\le x}\mu(d)\log(x/d) = \int_1^x M(u)\,du/u$ by partial summation. For $d > x$ we have $0 < x/d < 1$, so $\operatorname{sinc}(x/d) > 0$. Inserting (4.6) gives (4.5). □

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Proposition 4.3 (Mellin transform of H)**.** *For $\operatorname{Re} s > 1$ the integral $\widehat{H}(s) = \int_1^\infty H(x)x^{-s-1}\,dx$ converges absolutely and*

$$\widehat{H}(s) = \frac{\widehat{F}(s)}{\zeta(s)}, \qquad \widehat{H}(s) = \frac{\pi}{s}\cot\Big(\frac{\pi s}{2}\Big) + \frac{A(s)}{\zeta(s)} \quad (1 < \operatorname{Re} s < 2). \tag{4.7}$$

Proof. By (4.2) and dyadic decomposition, $\int_1^\infty |H|x^{-\sigma-1} < \infty$ for $\sigma > 1$. Moreover, by the substitution $x = du$, the sum over d of $\int_d^\infty |F(x/d)|x^{-\sigma-1}dx$ equals $\zeta(\sigma)\int_1^\infty |F(u)|u^{-\sigma-1}du < \infty$, so Fubini's theorem applies, and $\int_d^\infty F(x/d)x^{-s-1}\,dx = d^{-s}\widehat{F}(s)$ gives $\widehat{H} = \widehat{F}\sum_d \mu(d)d^{-s} = \widehat{F}/\zeta$. The second form is (2.9); it is stated for $1 < \operatorname{Re} s < 2$, where A is defined by (2.7), and only this range is used below. □

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Theorem 4.4 (L^2 criterion)**.** *The following are equivalent.* (a) *RH.* (b) $\int_1^\infty |H(x)|^2 x^{-2\sigma-1}\,dx < \infty$ *for every $\sigma > \frac12$.* (c) *For every $\varepsilon > 0$ there is K_ε with $\int_1^X H(x)^2\,dx \le K_\varepsilon X^{2+\varepsilon}$ for all $X \ge 1$.*

Proof. (b)⇒(c). With $\sigma = \frac{1}{2} + \varepsilon/2$, $\int_1^X H^2 \le X^{2\sigma+1} \int_1^X |H|^2 x^{-2\sigma-1} \le K_\varepsilon X^{2+\varepsilon}$.

(c)⇒(a). Let $\sigma > \frac{1}{2}$ and $\varepsilon = \sigma - \frac{1}{2}$. By Cauchy–Schwarz on $[2^q, 2^{q+1}]$, $q \ge 0$ an integer,

$$\int_{2^q}^{2^{q+1}} |H| x^{-\sigma-1}\,dx \le 2^{-q(\sigma+1)} 2^{q/2} \big(K_\varepsilon 2^{(q+1)(2+\varepsilon)}\big)^{1/2} \ll_\sigma 2^{-q(\sigma-1/2-\varepsilon/2)} = 2^{-q\varepsilon/2}, \tag{4.8}$$

which is summable. Hence $\widehat{H}(s)$ converges absolutely and locally uniformly on $\operatorname{Re} s > \frac{1}{2}$ and is holomorphic there. By (4.7) and uniqueness of continuation, $\frac{\pi}{s}\cot(\frac{\pi s}{2}) + A(s)/\zeta(s)$ extends holomorphically to $\frac{1}{2} < \operatorname{Re} s < 2$. The first term is holomorphic on $0 < \operatorname{Re} s < 2$. So A/ζ is holomorphic on $\frac{1}{2} < \operatorname{Re} s < 1$. If $\zeta(\rho) = 0$ with $\frac{1}{2} < \operatorname{Re}\rho < 1$, then $A(\rho) \ne 0$ by Theorem 3.1, and A/ζ has a pole at ρ, a contradiction. Since $\zeta(s) \ne 0$ for $\operatorname{Re} s > 1$ (Euler product) and on $\operatorname{Re} s = 1$ [5, §3.1, p. 45], RH holds.

(a)⇒(b). Assume RH and fix $\sigma_0 \in (\frac{1}{2}, 1)$. By (4.7), $\widehat{H}$ continues holomorphically to $\frac{1}{2} < \operatorname{Re} s < 2$: the cotangent term has no poles there, ζ has no zeros there, and at $s = 1$ the pole of ζ makes A/ζ vanish. Let $s = \sigma + it$ with $\sigma_0 \le \sigma \le \frac{3}{2}$ and $|t| \ge 2$. Then $|\cot(\pi s/2)| \le \coth(\pi|t|/2) \le \coth\pi$, since $|\cot z|^2 = (\cos^2 \operatorname{Re} z + \sinh^2 \operatorname{Im} z)/(\sin^2 \operatorname{Re} z + \sinh^2 \operatorname{Im} z)$. Next, $|A(s)| \ll \log|t|/|t|$: indeed $|2k - s| \ge |t|$ for all k, $|2k - s| \ge 2k - \frac{3}{2} \ge k/2$ for $k \ge 1$, so $\sum_{k \le |t|} \zeta(2)/(k|t|) + \sum_{k > |t|} 2\zeta(2)/k^2 \ll \log|t|/|t|$. Finally $1/\zeta(s) \ll_\varepsilon |t|^\varepsilon$ uniformly for $\sigma_0 \le \sigma \le 1$ by [5, Theorem 14.2, (14.2.1), p. 336, and (14.2.6), p. 337], and $1/\zeta(s) \ll \log^7|t|$ for $1 \le \sigma \le \frac{3}{2}$ by [5, (3.6.5), p. 51]. Hence

$$|\widehat{H}(\sigma + it)| \ll_\varepsilon |t|^{-1+\varepsilon} \qquad (\sigma_0 \le \sigma \le \tfrac{3}{2},\ |t| \ge 2), \tag{4.9}$$

and $\widehat{H}$ is bounded on the compact rectangle $[\sigma_0, \frac{3}{2}] \times [-2, 2]$. So $\sup_{\sigma_0 < \sigma \le 3/2} \int |\widehat{H}(\sigma + it)|^2 dt < \infty$. For $\sigma \ge \frac{3}{2}$ the function $\Upsilon_\sigma(v) = H(e^v)e^{-\sigma v}$ lies in $L^1 \cap L^2(0,\infty)$ by (4.2), $\widehat{H}(\sigma + it) = \int_0^\infty \Upsilon_\sigma(v) e^{-itv} dv$, and Plancherel's theorem [3, Theorem 9.13, p. 186] gives $\int |\widehat{H}(\sigma + it)|^2 dt = 2\pi \int_0^\infty |\Upsilon_\sigma|^2 \le 2\pi \int_0^\infty |\Upsilon_{3/2}|^2$. Thus

$$\sup_{\sigma > \sigma_0} \frac{1}{2\pi} \int_{-\infty}^{\infty} |\widehat{H}(\sigma + it)|^2\,dt < \infty. \tag{4.10}$$

Put $\mathcal{P}(z) = \widehat{H}(\sigma_0 - iz)$ for $\operatorname{Im} z > 0$, so that $\mathcal{P}(\xi + i\upsilon) = \widehat{H}(\sigma_0 + \upsilon - i\xi)$. By (4.10) and the Paley–Wiener theorem [3, Theorem 19.2, pp. 372–373] there is $\Psi \in L^2(0,\infty)$ with $\mathcal{P}(z) = \int_0^\infty \Psi(v) e^{ivz}\,dv$ for $\operatorname{Im} z > 0$. For $\upsilon > \frac{3}{2} - \sigma_0$ also $\mathcal{P}(\xi + i\upsilon) = \int_0^\infty H(e^v) e^{-\sigma_0 v} e^{-\upsilon v} e^{i\xi v}\,dv$. Both $\Psi(v)e^{-\upsilon v}$ and $H(e^v)e^{-(\sigma_0+\upsilon)v}$ lie in $L^1(0,\infty)$ and have the same Fourier transform, so they agree almost everywhere [3, Theorem 9.12, p. 185]. Hence $\int_0^\infty |H(e^v)|^2 e^{-2\sigma_0 v} dv = \int_0^\infty |\Psi|^2 < \infty$, which is $\int_1^\infty |H(x)|^2 x^{-2\sigma_0 - 1} dx < \infty$. Since $\sigma_0 \in (\frac{1}{2}, 1)$ was arbitrary and the integral decreases in σ, (b) holds. □

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Remark 4.5. A pointwise bound $H(x) \ll x^{1/2+\varepsilon}$ fails: by Proposition 4.1(ii), $H(x) + \mu(n)\log(n - x)$ has a finite limit as $x \to n^-$, so H is unbounded near every squarefree $n \ge 2$. This is why the pointwise criterion is stated for the primitive W in the next section.

5. The Clausen sum W

For $X \ge 1$ let

$$W(X) = \sum_{d \le X} \mu(d)\, d\, \mathrm{Cl}_2\Big(\frac{2\pi X}{d}\Big). \tag{5.1}$$

The term d vanishes at $X = d$ because $\mathrm{Cl}_2(2\pi) = 0$, so W is continuous, $W(1) = 0$, and by (2.2), $|W(X)| \le \mathrm{Cl}_2(\pi/3) X(X+1)/2$.

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Proposition 5.1. (i) $W(X) = -2\pi \int_1^X H(x)\,dx$ *for* $X \geq 1$. (ii) *With* $\mathcal{K}(v) = \mathrm{Cl}_2(2\pi v) + 2\pi v\, F(v)$ *for* $v \notin \mathbb{Z}$,

$$W(X) = -\int_1^X M(y)\,\mathcal{K}\Big(\frac{X}{y}\Big)\,dy. \tag{5.2}$$

(iii) *For* $\operatorname{Re} s > 2$ *the integral* $\widehat{W}(s) = \int_1^\infty W(X) X^{-s-1} dX$ *converges absolutely and*

$$\widehat{W}(s) = -\frac{2\pi}{s}\,\widehat{H}(s-1) = -\frac{2\pi}{s}\,\frac{\widehat{F}(s-1)}{\zeta(s-1)}. \tag{5.3}$$

The right-hand side is meromorphic on $\operatorname{Re} s > 0$ *and holomorphic in a neighbourhood of the half-line* $(1, \infty)$.

Proof. (i) For $X \notin \mathbb{Z}$, $\frac{d}{dX}\big(d\,\mathrm{Cl}_2(2\pi X/d)\big) = -2\pi F(X/d)$ by (2.1). On each open interval $(n, n+1)$, $n \geq 1$ an integer, every x/d is a non-integer, so W is continuously differentiable there with $W' = -2\pi H$. Since W is continuous on $[n, n+1]$ and H is integrable there by (4.2), $W(n+1) - W(n) = -2\pi \int_n^{n+1} H$, and also for intermediate endpoints; with $W(1) = 0$ this gives (i).

(ii) Write $d\,\mathrm{Cl}_2(2\pi X/d) = X\mathcal{Q}(X/d)$ with $\mathcal{Q}(v) = \mathrm{Cl}_2(2\pi v)/v$. Partial summation, with $M(1^-) = 0$ and $\mathcal{Q}(1) = 0$, gives $\sum_{d \leq X} \mu(d)\mathcal{Q}(X/d) = \int_1^X M(y)\,(X/y^2)\mathcal{Q}'(X/y)\,dy$. Since $v^2\mathcal{Q}'(v) = 2\pi v\,\mathrm{Cl}_2'(2\pi v) - \mathrm{Cl}_2(2\pi v) = -\mathcal{K}(v)$, (ii) follows.

(iii) Absolute convergence follows from $|W(X)| \ll X^2$. Integrating by parts with (i), the boundary terms vanish for $\operatorname{Re} s > 2$, and $\widehat{W}(s) = \frac{1}{s}\int_1^\infty W'(X) X^{-s} dX = -\frac{2\pi}{s}\widehat{H}(s-1)$. By Theorem 2.2, $\widehat{F}$ is holomorphic on $\operatorname{Re} s > -1$, so $\widehat{F}(s-1)/\zeta(s-1)$ is meromorphic on $\operatorname{Re} s > 0$. On the real half-line $s - 1 > 0$, $\zeta(s-1) \neq 0$: for $0 < \sigma < 1$, $(1 - 2^{1-\sigma})\zeta(\sigma) = \sum_n (-1)^{n-1} n^{-\sigma} > 0$, and $\zeta(\sigma) > 0$ for $\sigma > 1$; at $s - 1 = 1$, $1/\zeta$ is holomorphic with a zero. Since the zeros of ζ are isolated, the right-hand side of (5.3) is holomorphic near $(1, \infty)$. □

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Lemma 5.2 (Landau)**.** *Let* f *be real, locally bounded and measurable on* $[1, \infty)$*, with* $f(X) \geq 0$ *for* $X \geq X_0$*. Suppose* $g(s) = \int_1^\infty f(X) X^{-s-1} dX$ *converges absolutely for* $\operatorname{Re} s > \sigma_1$*, and that* g *extends holomorphically to a neighbourhood of the real segment* $[\sigma_2, \sigma_1]$*, where* $\sigma_2 < \sigma_1$*. Then the integral converges absolutely for* $\operatorname{Re} s > \sigma_2$.

Proof. Let σ_c be the infimum of the real σ with $\int_1^\infty |f| X^{-\sigma-1} < \infty$; then $\sigma_c \leq \sigma_1$, and g is holomorphic on $\operatorname{Re} s > \sigma_c$. Suppose $\sigma_c > \sigma_2$. Then g is holomorphic on a neighbourhood U of $[\sigma_c, \infty)$, the union of the half-plane and the given neighbourhood. Choose $h_0 > 0$ so small that the closed disc of radius $2h_0$ about $\sigma_3 = \sigma_c + h_0$ lies in U. Splitting off $[1, X_0]$, where f is bounded, we may assume $f \geq 0$ on $[1, \infty)$. The extension agrees with the integral on the part of the disc in $\operatorname{Re} s > \sigma_c$, because that part is convex and contains real points of (σ_c, ∞), where both are equal by uniqueness of continuation. For $k \geq 0$, $g^{(k)}(\sigma_3) = \int_1^\infty f(X)(-\log X)^k X^{-\sigma_3-1} dX$ by differentiation under the integral. Taylor's theorem at σ_3 gives, for $h_0 < h < 2h_0$,

$$g(\sigma_3 - h) = \sum_{k \geq 0} \frac{h^k}{k!} \int_1^\infty f(X)(\log X)^k X^{-\sigma_3-1}\,dX = \int_1^\infty f(X) X^{-(\sigma_3-h)-1}\,dX, \tag{5.4}$$

where Tonelli's theorem applies because every term is nonnegative ($X \geq 1$). The left side is finite, so the integral converges at the real point $\sigma_3 - h < \sigma_c$, a contradiction. Hence $\sigma_c \leq \sigma_2$. □

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Theorem 5.3 (Pointwise and one-sided criteria)**.** *The following are equivalent.* (a) *RH.* (b) *For every* $\varepsilon > 0$ *there is* K_ε *with* $|W(X)| \leq K_\varepsilon X^{3/2+\varepsilon}$ *for* $X \geq 1$. (c) *For every* $\varepsilon > 0$ *there is* K_ε *with* $W(X) \leq K_\varepsilon X^{3/2+\varepsilon}$ *for* $X \geq 1$. (d) *For every* $\varepsilon > 0$ *there is* K_ε *with* $W(X) \geq -K_\varepsilon X^{3/2+\varepsilon}$ *for* $X \geq 1$.

Proof. (a)⇒(b). Under RH, $M(y) \ll_\varepsilon y^{1/2+\varepsilon}$ [5, Theorem 14.25(C), p. 370]. In (5.2) substitute $y = X/v$:

$$|W(X)| \le \int_1^X \Big|M\Big(\frac{X}{v}\Big)\Big||\mathcal{K}(v)|\frac{X}{v^2}\,dv \ll_\varepsilon X^{3/2+\varepsilon}\int_1^\infty v^{-5/2}|\mathcal{K}(v)|\,dv. \tag{5.5}$$

Since $|\mathcal{K}(v)| \le 1.015 + 2\pi v|F(v)|$ and $\int_n^{n+1}|F| = \int_0^1|F|$, the last integral is finite.

(b)⇒(c) and (b)⇒(d) are trivial.

(c)⇒(a). Fix $\varepsilon > 0$ and put $\beta = \frac{3}{2} + \varepsilon$ and $f(X) = K_\varepsilon X^\beta - W(X) \ge 0$. Its Mellin integral $g(s) = K_\varepsilon/(s-\beta) - \widehat{W}(s)$ converges absolutely for $\operatorname{Re} s > 2$, and by Proposition 5.1(iii) it extends holomorphically to a neighbourhood of every real segment $[\sigma_2, 2]$ with $\beta < \sigma_2$. By Lemma 5.2 the integral of f converges absolutely for $\operatorname{Re} s > \beta$, hence so does $\widehat{W}$, since $|W| \le f + K_\varepsilon X^\beta$. Thus $-\frac{2\pi}{s}\widehat{F}(s-1)/\zeta(s-1)$ is holomorphic on $\operatorname{Re} s > \beta$, i.e. $\widehat{F}(w)/\zeta(w) = \frac{\pi}{w}\cot\frac{\pi w}{2} + A(w)/\zeta(w)$ is holomorphic on $\operatorname{Re} w > \frac{1}{2} + \varepsilon$. The cotangent term is holomorphic on $0 < \operatorname{Re} w < 2$, so A/ζ is holomorphic on $\frac{1}{2} + \varepsilon < \operatorname{Re} w < 1$. A zero ρ of ζ in that strip would be a pole of A/ζ, because $A(\rho) \ne 0$ by Theorem 3.1. So there is none, and with $\zeta(s) \ne 0$ on $\operatorname{Re} s \ge 1$ [5, §3.1, p. 45] and $\varepsilon \to 0$ this gives RH.

(d)⇒(a). The same, with $f(X) = W(X) + K_\varepsilon X^\beta$ and $g(s) = \widehat{W}(s) + K_\varepsilon/(s-\beta)$. □

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Theorem 5.4 ($\Omega_\pm$ bounds)**.** *Let $\rho = \frac{1}{2} + i\gamma$, $\gamma > 0$, be a simple zero of ζ, and*

$$r_\rho = -\frac{2\pi}{1+\rho}\,\frac{A(\rho)}{\zeta'(\rho)} \ne 0. \tag{5.6}$$

Then $\limsup_{X\to\infty} W(X)X^{-3/2} \ge |r_\rho|$ and $\liminf_{X\to\infty} W(X)X^{-3/2} \le -|r_\rho|$. For the first zero, $\gamma_1 = 14.13472514\ldots$, which lies on the critical line and is simple (classical numerical facts, used here as inputs), numerically $|\zeta'(\rho_1)| = 0.79316$ and $|r_{\rho_1}| = 0.0821128$.

Proof. $r_\rho \ne 0$ by Theorem 3.1, and r_ρ is the residue of $\widehat{W}$ at $s = 1+\rho$ by (5.3), since $\widehat{F}(\rho) = A(\rho)$ by (2.9). Suppose there are $b < |r_\rho|$ and X_0 with $W(X) \le bX^{3/2}$ for $X \ge X_0$. Let $f(X) = bX^{3/2} - W(X)$ and $g(s) = b/(s-\frac{3}{2}) - \widehat{W}(s)$. The function f is continuous and $f \ge 0$ on $[X_0,\infty)$; its Mellin integral g converges absolutely for $\operatorname{Re} s > 2$ because $|W(X)| \ll X^2$; and g extends holomorphically to a neighbourhood of $[\sigma_2, 2]$ for every $\sigma_2 > \frac{3}{2}$ by Proposition 5.1(iii). Lemma 5.2 gives absolute convergence of $\int_1^\infty fX^{-s-1}$ on $\operatorname{Re} s > \frac{3}{2}$; hence g is holomorphic there, and g is meromorphic near $s_0 = \frac{3}{2} + i\gamma$ with $g(s) = -r_\rho/(s-s_0) + O(1)$. Since f is real, $g(\bar s) = \overline{g(s)}$. For real $\sigma > \frac{3}{2}$ and real ϕ_0,

$$g(\sigma) + \tfrac{1}{2}e^{i\phi_0}g(\sigma+i\gamma) + \tfrac{1}{2}e^{-i\phi_0}g(\sigma-i\gamma) = \int_1^\infty f(X)X^{-\sigma-1}\big(1+\cos(\gamma\log X - \phi_0)\big)\,dX \ge -Q_0, \tag{5.7}$$

with $Q_0 = 2\int_1^{X_0}|f|X^{-5/2}dX$, because the integrand is nonnegative on $[X_0,\infty)$. As $\sigma \to \frac{3}{2}^+$, $\widehat{W}(\sigma)$ stays bounded by Proposition 5.1(iii), so $g(\sigma) = b/(\sigma-\frac{3}{2}) + O(1)$, while $g(\sigma \pm i\gamma) = -r_\rho^{\pm}/(\sigma - \frac{3}{2}) + O(1)$ with $r_\rho^+ = r_\rho$, $r_\rho^- = \overline{r_\rho}$. The left side of (5.7) is therefore $(b - \operatorname{Re}(e^{i\phi_0}r_\rho))/(\sigma-\frac{3}{2}) + O(1)$. Choosing $e^{i\phi_0}r_\rho = |r_\rho|$ it tends to $-\infty$, a contradiction. Hence $\limsup W/X^{3/2} \ge |r_\rho|$. For the $\liminf$ use $f = W + bX^{3/2}$, $g = \widehat{W} + b/(s-\frac{3}{2})$ and $e^{i\phi_0}r_\rho = -|r_\rho|$. The numerical values come from `route_constants.py` (mpmath, 30 digits); mpmath locates ρ_1 on the critical line by assumption, so the location of ρ_1 is an input and not a consequence of this computation; the computed value $\zeta'(\rho_1) \ne 0$ agrees with the simplicity of ρ_1. □

6. The $d = 2$ transfer and the corner

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Lemma 6.1. *Let $U \ge 1$ and $\mu_U(n) = \mu(n)$ for $n \le U$, $\mu_U(n) = 0$ for $n > U$. For $1 \le n \le U^2$,*

$$\mu(n) = 2\mu_U(n) - (\mu_U * \mu_U * 1)(n). \tag{6.1}$$

Proof. Let $\mu^U = \mu - \mu_U$, and let $\mathbf{1}_{\{1\}}$ be the unit of Dirichlet convolution, so that $\mu * 1 = \mathbf{1}_{\{1\}}$. Then $\mu^U * 1 * \mu^U = \mu - 2\mu_U + \mu_U * \mu_U * 1$. A nonzero term of $(\mu^U * \mu^U)(n)$ needs $n = d_1 d_2$ with $d_1, d_2 > U$, so $n > U^2$; hence $(\mu^U * 1 * \mu^U)(n) = 0$ for $n \le U^2$. This is the Möbius form of the device behind Vaughan's identity for Λ [2, Ch. 24, pp. 138–139]. □

For $Y \ge 1$ let

$$G(Y) = \sum_{\ell \le Y} \ell \,\mathrm{Cl}_2\Big(\frac{2\pi Y}{\ell}\Big), \qquad E_G(Y) = G(Y) - aY^2, \tag{6.2}$$

and for $\operatorname{Re} s > 0$ let

$$\Theta(s) = \int_1^\infty \mathrm{Cl}_2(2\pi v)\, v^{-s-1}\, dv = -\frac{2\pi}{s}\,\widehat{F}(s-1). \tag{6.3}$$

The second equality holds for $\operatorname{Re} s > 1$ by parts, using $\mathrm{Cl}_2(2\pi) = 0$ and (2.1), and then for $\operatorname{Re} s > 0$ by continuation; $\Theta(2) = -\pi\widehat{F}(1) = a$ by (2.23). G is continuous for the same reason as W, $G(1) = 0$, and $|G(Y)| \le \mathrm{Cl}_2(\pi/3)Y(Y+1)/2$.

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Lemma 6.2. (i) *For* $\operatorname{Re} s > 2$, $\widehat{G}(s) = \int_1^\infty G(Y)Y^{-s-1}dY = \zeta(s-1)\Theta(s)$. (ii) *For every* $\delta \in (0,1)$ *there is* c_δ *with* $|E_G(Y)| \le c_\delta Y^{1+\delta}$ *for* $Y \ge 1$. (iii) *For* $Y \notin \mathbb{Z}$, $E_G'(Y) = -2\pi R(Y)$, *where* $R(Y) = \sum_{\ell \le Y} F(Y/\ell) - (\log 2 - C)Y$.

Proof. (i) By Fubini's theorem, justified by Tonelli's theorem applied to the absolute values, and the substitution $Y = \ell v$, $\int_\ell^\infty \ell\,\mathrm{Cl}_2(2\pi Y/\ell)Y^{-s-1}dY = \ell^{1-s}\Theta(s)$, and $\sum_\ell \ell^{1-\operatorname{Re} s} < \infty$.

(ii) Put $w = s - 1$, $s = \sigma + it$. By (6.3) and (2.9), for $0 < \operatorname{Re} w < 2$,

$$\widehat{G}(s) = -\frac{2\pi}{s}\,\zeta(w)\Big(\frac{\pi}{w}\cot\frac{\pi w}{2}\,\zeta(w) + A(w)\Big). \tag{6.4}$$

For $|t| \ge 2$ and $0 < \operatorname{Re} w \le \frac{3}{2}$ we have $|\frac{\pi}{w}\cot\frac{\pi w}{2}| \le \pi\coth(\pi)/|t|$, because $|w| \ge |t|$ and $|\cot z| \le \coth|\operatorname{Im} z|$; and $|A(w)| \ll \log|t|/|t|$, because $|2k - w| \ge |t|$ and $|2k - w| \ge 2k - \frac{3}{2} \ge k/2$ for $k \ge 1$, so that the series in (2.7) is at most $\sum_{k \le |t|} \zeta(2)/(k|t|) + \sum_{k > |t|} 2\zeta(2)/k^2$. Hence

$$|\widehat{G}(s)| \ll |t|^{-2}(|\zeta(w)|^2 + |\zeta(w)|\log|t|) \qquad (1 < \sigma \le \tfrac{5}{2},\ |t| \ge 2). \tag{6.5}$$

On $\sigma = \frac{5}{2}$, $|\zeta(w)| \le \zeta(\frac{3}{2})$, so $\widehat{G}(\frac{5}{2} + it) \in L^1(dt)$. The function $Y \mapsto G(Y)Y^{-5/2}$, extended by 0 to $(0,1)$, is continuous, and in the variable $v = \log Y$ it is integrable with integrable Fourier transform; the inversion theorem [3, Theorem 9.11, p. 185] gives

$$G(Y) = \frac{1}{2\pi i}\int_{(5/2)} \widehat{G}(s)\, Y^s\, ds \qquad (Y \ge 1). \tag{6.6}$$

Move the line to $\sigma = 1 + \delta$. In $1 + \delta \le \sigma \le \frac{5}{2}$ the only pole of $\widehat{G}$ is $s = 2$, from $\zeta(s-1)$, with residue $\Theta(2)Y^2 = aY^2$; Θ is holomorphic on $\operatorname{Re} s > 0$. On the horizontal segments $\sigma + iT$, $1 + \delta \le \sigma \le \frac{5}{2}$, Theorem 4.11 of [5, p. 77], applied with a partial sum of length $|T|$, gives $|\zeta(w)| \ll_\delta |T|^{1-\delta}$ uniformly for $\operatorname{Re} w \ge \delta$, so by (6.5) $|\widehat{G}(\sigma + iT)| \ll |T|^{-2\delta}\log|T| \to 0$. On the line $\sigma = 1 + \delta$, $\operatorname{Re} w = \delta$ and the convexity bound $\zeta(\delta + it) \ll_\varepsilon |t|^{(1-\delta)/2+\varepsilon}$ [5, (5.1.4), p. 96] with $\varepsilon = \delta/4$ gives $|\widehat{G}(1 + \delta + it)| \ll |t|^{-1-\delta/2}$ for $|t| \ge 2$; near $t = 0$, $\widehat{G}$ is continuous on the line. Hence

$$E_G(Y) = \frac{1}{2\pi i}\int_{(1+\delta)} \widehat{G}(s)\, Y^s\, ds, \qquad |E_G(Y)| \le \frac{Y^{1+\delta}}{2\pi}\int_{-\infty}^{\infty} |\widehat{G}(1 + \delta + it)|\, dt = c_\delta Y^{1+\delta}. \tag{6.7}$$

(iii) On $(n, n+1)$, $n \ge 1$ an integer, each term of (6.2) is differentiable and $\frac{d}{dY}(\ell\,\mathrm{Cl}_2(2\pi Y/\ell)) = -2\pi F(Y/\ell)$ by (2.1), so $G'(Y) = -2\pi\sum_{\ell \le Y} F(Y/\ell)$; with $\frac{d}{dY}aY^2 = 2aY = 2\pi(C - \log 2)Y$ this gives $E_G' = -2\pi R$. □

For $X \geq 1$ define

$$W^\flat(X) = \sum_{n \leq \sqrt{X}} \mu(n)\, n\, \mathrm{Cl}_2\Big(\frac{2\pi X}{n}\Big), \tag{6.8}$$

$$\mathcal{R}(X) = \sum_{\substack{d,j \leq \sqrt{X} \\ dj \leq \sqrt{X}}} \mu(d)\mu(j)\, dj\, E_G\Big(\frac{X}{dj}\Big), \qquad \mathcal{C}(X) = \sum_{\substack{d,j \leq \sqrt{X} \\ dj > \sqrt{X}}} \mu(d)\mu(j)\, dj\, E_G\Big(\frac{X}{dj}\Big). \tag{6.9}$$

In $\mathcal{C}(X)$ the argument $Y = X/(dj)$ lies in $[1, \sqrt{X})$; in $\mathcal{R}(X)$ it lies in $[\sqrt{X}, X]$.

Reviewed by Claude (review 1)

Proposition 6.3 (Exact decomposition)**.** *For $X \geq 1$,*

$$W(X) = 2W^\flat(X) - aX^2 m(\sqrt{X})^2 - \mathcal{R}(X) - \mathcal{C}(X). \tag{6.10}$$

Moreover $|2W^\flat(X)| \leq \mathrm{Cl}_2(\pi/3)(X + \sqrt{X})$, and for every $\delta \in (0,1)$,

$$|\mathcal{R}(X)| \leq c_\delta X^{3/2+\delta/2}(1 + \log\sqrt{X})^2. \tag{6.11}$$

Proof. By Lemma 6.1 with $U = \sqrt{X}$, for $n \leq X$, $\mu(n) = 2\mu_U(n) - \sum_{dj\ell = n,\ d,j \leq U} \mu(d)\mu(j)$. Hence

$$\begin{aligned} W(X) &= 2W^\flat(X) - \sum_{d,j \leq \sqrt{X}} \mu(d)\mu(j) \sum_{\ell \leq X/(dj)} dj\ell\, \mathrm{Cl}_2\Big(\frac{2\pi X}{dj\ell}\Big) \\ &= 2W^\flat(X) - \sum_{d,j \leq \sqrt{X}} \mu(d)\mu(j)\, dj\, G\Big(\frac{X}{dj}\Big), \end{aligned} \tag{6.12}$$

where $dj \leq X$, so $X/(dj) \geq 1$. Inserting $G(Y) = aY^2 + E_G(Y)$, the main terms give aX^2 times the double sum of $\mu(d)\mu(j)/(dj)$ over $d, j \leq \sqrt{X}$, which is $aX^2 m(\sqrt{X})^2$, and the error terms split into $\mathcal{R} + \mathcal{C}$. The bound on $W^\flat$ follows from (2.2) and $2\sum_{n \leq \sqrt{X}} n \leq \sqrt{X}(\sqrt{X}+1)$. For (6.11), the number of pairs with $dj = n$ is at most $\tau(n)$, so by Lemma 6.2(ii)

$$|\mathcal{R}(X)| \leq c_\delta \sum_{n \leq \sqrt{X}} \tau(n)\, n \Big(\frac{X}{n}\Big)^{1+\delta} \leq c_\delta X^{1+\delta}(\sqrt{X})^{1-\delta} \sum_{n \leq \sqrt{X}} \frac{\tau(n)}{n} = c_\delta X^{3/2+\delta/2} \sum_{n \leq \sqrt{X}} \frac{\tau(n)}{n}, \tag{6.13}$$

and $\sum_{n \leq y} \tau(n)/n \leq (\sum_{n \leq y} 1/n)^2 \leq (1 + \log y)^2$. □

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Theorem 6.4 (Reduction of RH to the corner)**.** *The following are equivalent.* (a) *RH.* (b) *For every $\varepsilon > 0$ there is K_ε with $\mathcal{C}(X) \geq -K_\varepsilon X^{3/2+\varepsilon}$ for all $X \geq 1$.* (c) *For every $\varepsilon > 0$ there is K_ε with $|\mathcal{C}(X)| \leq K_\varepsilon X^{3/2+\varepsilon}$ for all $X \geq 1$.*

Proof. (b)⇒(a). By (6.10), $a > 0$ and Proposition 6.3,

$$\begin{aligned} W(X) &= 2W^\flat(X) - aX^2 m(\sqrt{X})^2 - \mathcal{R}(X) - \mathcal{C}(X) \\ &\leq \mathrm{Cl}_2(\tfrac{\pi}{3})(X + \sqrt{X}) + c_\delta X^{3/2+\delta/2}(1 + \log\sqrt{X})^2 + K_\varepsilon X^{3/2+\varepsilon}. \end{aligned} \tag{6.14}$$

With $\delta = \min(\varepsilon, \frac{1}{2})$ the right side is at most $K'_\varepsilon X^{3/2+\varepsilon}$ for $X \geq 1$, so condition (c) of Theorem 5.3 holds for every $\varepsilon > 0$, and RH follows. No information on $m(\sqrt{X})$ is used: the term $-aX^2 m(\sqrt{X})^2$ is ≤ 0 and is dropped.

(a)⇒(c). Assume RH. By Theorem 5.3, $W \ll X^{3/2+\varepsilon}$. By partial summation, for $\mathrm{Re}\, s > 1$, $\sum_n \mu(n) n^{-s} = s\int_1^\infty M(v) v^{-s-1} dv$; under RH, $M(v) \ll v^{1/2+\varepsilon}$ [5, Theorem 14.25(C), p. 370], so the integral converges absolutely and uniformly for $s \in [1,2]$, and letting $s \to 1^+$ gives $\int_1^\infty M(v) v^{-2} dv = \lim_{s \to 1^+} 1/(s\zeta(s)) = 0$. Then

$$m(y) = \frac{M(y)}{y} + \int_1^y \frac{M(v)}{v^2}\, dv = \frac{M(y)}{y} - \int_y^\infty \frac{M(v)}{v^2}\, dv \ll_\varepsilon y^{-1/2+\varepsilon} \qquad (y \geq 1), \tag{6.15}$$

so $aX^2 m(\sqrt{X})^2 \ll X^{3/2+\varepsilon}$. Solving (6.10) for $\mathcal{C}$ and using Proposition 6.3 gives (c).

(c)⇒(b) is trivial. □

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Remark 6.5. $\mathcal{C}(X)$ depends on μ only through its values on $[1,\sqrt{X}]$, and on E_G only on $[1,\sqrt{X})$. The function E_G involves no Möbius values. By Lemma 6.2(iii), the derivative of E_G is $-2\pi R$, and at each integer $n\ge 2$ the function R has logarithmic singularities, $(\tau(n)-1)\log|Y-n|$ from the left and $\tau(n)\log|Y-n|$ from the right, coming from the terms $\ell\mid n$ of $\sum_\ell F(Y/\ell)$. Every attempt recorded in Section 12 stopped at the size of $\sum|\text{terms}|$ of $\mathcal{C}$, which grows like X^2 in the measured range (Section 12).

7. Other exact identities

Reviewed by Claude (review 1)

Proposition 7.1 (Recursion over large primes)**.** *For $X\ge 1$ let $W^{\mathrm{sm}}(X)$ be the sum* (5.1) *restricted to n whose prime factors are all $\le\sqrt{X}$. Then*

$$W(X)=W^{\mathrm{sm}}(X)-\sum_{\sqrt{X}<p\le X} p\,W\Big(\frac{X}{p}\Big). \tag{7.1}$$

Proof. A squarefree $n\le X$ has at most one prime factor $p>\sqrt{X}$, and then $n=pb$ with $b\le X/p<\sqrt{X}<p$, so $p\nmid b$ and $\mu(pb)=-\mu(b)$. Conversely each pair (p,b) with $\sqrt{X}<p\le X$, $b\le X/p$, gives such an n, and $X/p\ge 1$. Hence the non-smooth part of (5.1) is $-\sum_p p\sum_{b\le X/p}\mu(b)\,b\,\mathrm{Cl}_2(2\pi(X/p)/b)=-\sum_p pW(X/p)$. □

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Proposition 7.2 (lcm form)**.** *Let $X\ge 1$, let $\mathcal{D}$ be the set of squarefree $d\le X$, let $\Xi_X(n)=n\,\mathrm{Cl}_2(2\pi X/n)$ for $n\le X$ and $\Xi_X(n)=0$ for $n>X$, and let $\mathbf{K}$ be the matrix $(\Xi_X([d,d']))_{d,d'\in\mathcal{D}}$, where $[d,d']$ is the least common multiple. Let $\boldsymbol{\mu}=(\mu(d'))_{d'\in\mathcal{D}}$ and let $\mathbf{e}_1$ be the coordinate vector of the index $1\in\mathcal{D}$. Then $\mathbf{K}\boldsymbol{\mu}=W(X)\,\mathbf{e}_1$. In particular $W(X)=\sum_{d,d'\in\mathcal{D}}\mu(d)\mu(d')\Xi_X([d,d'])$, and $W(X)=1/(\mathbf{K}^{-1})_{11}$ whenever $\mathbf{K}$ is invertible.*

Proof. Fix $\bar d\in\mathcal{D}$. Each squarefree d' factors uniquely as $d'=d_1d_2$ with $d_1\mid\bar d$, $(d_2,\bar d)=1$, and then $[\bar d,d']=\bar d d_2$ and $\mu(d')=\mu(d_1)\mu(d_2)$. Every squarefree d' with $\Xi_X([\bar d,d'])\ne 0$ has $d'\le[\bar d,d']\le X$, so it lies in $\mathcal{D}$. Thus $(\mathbf{K}\boldsymbol{\mu})_{\bar d}=\sum_{d_1\mid\bar d}\mu(d_1)\sum_{(d_2,\bar d)=1}\mu(d_2)\Xi_X(\bar d d_2)$. The inner sum does not depend on d_1, and $\sum_{d_1\mid\bar d}\mu(d_1)=0$ for $\bar d>1$. For $\bar d=1$ the sum is $\sum_{d'\in\mathcal{D}}\mu(d')\Xi_X(d')=W(X)$. Taking the inner product with $\boldsymbol{\mu}$ gives the quadratic form, because $\mu(1)=1$. If $\mathbf{K}$ is invertible, $\boldsymbol{\mu}=W(X)\mathbf{K}^{-1}\mathbf{e}_1$, and the coordinate of the index 1 gives $1=W(X)(\mathbf{K}^{-1})_{11}$. □

Invertibility of $\mathbf{K}$ is not proved. Numerically, for $X=300.5$, 1000.5, 2500.5, the identity $\mathbf{K}\boldsymbol{\mu}=W(X)\mathbf{e}_1$ holds to 10^{-11}, $\det\mathbf{K}/\det\mathbf{K}^{(1)}$ reproduces $W(X)$, and the condition numbers are $4.9\cdot10^3$, $5.8\cdot10^{14}$, $8.9\cdot10^{14}$; the inertia of $\mathbf{K}^{(1)}$ is close to balanced (84/98, 281/326, 695/827 positive/negative eigenvalues), and $\boldsymbol{\mu}$ has almost no weight on the eigenvectors with eigenvalues of large modulus (the matrix $\mathbf{K}^{(1)}$ is $\mathbf{K}$ without the row and column of the index 1).

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Proposition 7.3 ($d=3$)**.** *Let $X\ge 1$, $U=X^{1/3}$, $G_2(Y)=\sum_{\ell_1\ell_2\le Y}\ell_1\ell_2\,\mathrm{Cl}_2(2\pi Y/(\ell_1\ell_2))$, and*

$$\begin{aligned} T_1&=\sum_{n\le U}\mu(n)\,n\,\mathrm{Cl}_2\Big(\frac{2\pi X}{n}\Big), \qquad T_2=\sum_{d,j\le U}\mu(d)\mu(j)\,dj\,G\Big(\frac{X}{dj}\Big),\\ T_3&=\sum_{d,j,h\le U}\mu(d)\mu(j)\mu(h)\,djh\,G_2\Big(\frac{X}{djh}\Big). \end{aligned} \tag{7.2}$$

Then $W(X)=3T_1-3T_2+T_3$.

Proof. Let $\mu^U=\mu-\mu_U$. A nonzero term of $(\mu^U)^{*3}(n)$ needs $n=d_1d_2d_3$ with $d_1,d_2,d_3>U$, so $n>U^3$; hence $(\mu^U)^{*3}*1^{*2}$ vanishes on $[1,U^3]$, and its expansion is $\mu-3\mu_U+3\mu_U^{*2}*1-\mu_U^{*3}*1^{*2}$, using $\mu*1=\mathbf{1}_{\{1\}}$. Multiply by $n\,\mathrm{Cl}_2(2\pi X/n)$ and sum over $n\le X$. □

The route of Theorem 6.4 does not use Proposition 7.3. In particular it needs no bound on $m(X^{1/3})$ or on any other tail of m; such quantities appear only if T_2 and T_3 are split into main terms and errors, and that split is not part of the route.

8. Two facts about methods

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Proposition 8.1. *Let $W_{\mu^2}(X) = \sum_{n\le X} \mu(n)^2\, n\, \mathrm{Cl}_2(2\pi X/n)$. For every $\delta \in (0, \frac12)$,*

$$W_{\mu^2}(X) = \frac{6}{\pi^2}\, a\, X^2 + O_\delta(X^{3/2+\delta}), \qquad \frac{6}{\pi^2} a = 0.0967299459812\ldots \tag{8.1}$$

Proof. By Fubini's theorem, justified as in Lemma 6.2(i) by $|\mathrm{Cl}_2| \le \mathrm{Cl}_2(\pi/3)$ and $\sum_n n^{1-\mathrm{Re}\, s} < \infty$, and the substitution $X = nv$, $\int_n^\infty n\, \mathrm{Cl}_2(2\pi X/n) X^{-s-1} dX = n^{1-s}\Theta(s)$, and $\sum_n \mu(n)^2 n^{-z} = \zeta(z)/\zeta(2z)$ for $\mathrm{Re}\, z > 1$, so $\int_1^\infty W_{\mu^2} X^{-s-1} dX = \Theta(s)\zeta(s-1)/\zeta(2s-2)$ for $\mathrm{Re}\, s > 2$. On $\mathrm{Re}\, s \ge \frac32 + \delta$, $|1/\zeta(2s-2)| \le \zeta(1+2\delta)$. With this extra bounded factor, the bound (6.5) holds for the new transform on $\frac32 + \delta \le \sigma \le \frac52$, $|t| \ge 2$; it is integrable on $\sigma = \frac52$, so the inversion formula (6.6) holds for W_{μ^2} by [3, Theorem 9.11, p. 185], since W_{μ^2} is continuous and $O(X^2)$. Move the line from $\frac52$ to $\frac32 + \delta$: the only pole is $s = 2$, with residue $\Theta(2)X^2/\zeta(2) = 6aX^2/\pi^2$; on the horizontal segments $|\zeta(w)| \ll |T|^{1/2}$ uniformly for $\mathrm{Re}\, w \ge \frac12$ [5, Theorem 4.11, p. 77], so the integrand is $\ll |T|^{-1} \log |T|$; on the line, $|\zeta(\frac12 + \delta + it)| \ll |t|^{1/4}$ by [5, (5.1.4), p. 96], so the integrand is $\ll |t|^{-3/2}$. □

Reviewed by Claude (review 1)

Remark 8.2 (What a bound for $\mathcal{C}$ must use). For squarefree d, d' one has $\mu(d)\mu(d') = \mu(dd'/(d,d')^2)$, and the same relation holds with μ replaced by μ^2, or by $n \mapsto \mu(n)^2 \prod_{p\mid n} \chi(p)$ for any χ from the primes to $\{\pm 1\}$. An argument that uses μ only through this relation, $|\mu| \le 1$ and the support on squarefree integers applies equally to μ^2. By Proposition 8.1, $W_{\mu^2}(X) \sim (6/\pi^2) a X^2$, so such an argument cannot give $W(X) \ll X^{3/2+\varepsilon}$. For the corner inequality (14.1) the analogous conclusion is not proved: the route from the corner to RH uses $\mu * 1 = \mathbf{1}_{\{1\}}$ through Lemma 6.1, which fails for μ^2, so a bound for the μ^2-corner $\sum \mu(d)^2\mu(j)^2 dj\, E_G(X/(dj))$ would not bound W_{μ^2}. The conclusion would follow if the μ^2-corner violated the one-sided bound; the measured values in Section 12 ($-0.00090X^2$, $-0.00121X^2$, $-0.00148X^2$ for the part with $X/(dj) \le 64$) are negative and grow like X^2, but this is not proved.

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Remark 8.3 (Localization, not proved). An inequality of the form $W(X) \le c_1 X + c_\varepsilon X^{3/2+\kappa+\varepsilon}(1 + J(\sqrt{X}))$, with $J(y) = \int_1^y M(v)^2 v^{-2} dv + M(y)^2/y$ and κ a subconvexity exponent for ζ on the critical line, was derived in exploratory work through a mean-value argument in $\log y$. Its proof has not been written out or checked, and nothing in this document uses it.

9. A Hilbert-space form: the Müntz transform of the log-sine kernel

Let

$$\mathfrak{k}(t) = F(1/t) = \log\Big|2\sin\frac{\pi}{t}\Big| \quad (0 < t \le 1,\ 1/t \notin \mathbb{Z}), \qquad \mathfrak{k}(t) = 0 \quad (t > 1). \tag{9.1}$$

For $\sigma > 0$ put

$$\mathcal{N}_\sigma = \int_1^\infty F(v)^2 v^{-2\sigma-1}\, dv \le \frac{\pi^2}{12} \sum_{n\ge 1} n^{-2\sigma-1} < \infty, \tag{9.2}$$

using $\int_n^{n+1} F^2 = \pi^2/12$ (proof of Theorem 2.2). The substitution $t = 1/v$ gives $\int_0^\infty |\mathfrak{k}(t)| t^{\sigma-1} dt = \int_1^\infty |F(v)| v^{-\sigma-1} dv < \infty$ and $\int_0^\infty \mathfrak{k}(t)^2 t^{2\sigma-1} dt = \mathcal{N}_\sigma$, and

$$\int_0^\infty \mathfrak{k}(t)\, t^{s-1}\, dt = \int_1^\infty F(v)\, v^{-s-1}\, dv = \widehat{F}(s) \qquad (\mathrm{Re}\, s > 0), \qquad \int_0^\infty \mathfrak{k} = \widehat{F}(1) = \log 2 - C. \tag{9.3}$$

Reviewed by Claude (review 1)

Proposition 9.1 (Dilation sums of $\mathfrak{k}$)**.** *Let $t > 0$ with $1/t \notin \mathbb{Z}$ and put $x = 1/t$. Then*

$$\sum_{n\ge1} \mathfrak{k}(nt) - \frac{1}{t}\int_0^\infty \mathfrak{k} = R(x), \qquad \sum_{n\ge1} \mu(n)\,\mathfrak{k}(nt) = H(x) \quad (x \ge 1), \tag{9.4}$$

with R as in Lemma 6.2(iii)*; both sums have the finitely many nonzero terms $n \le x$, and for $x < 1$ the second sum is empty. For $\operatorname{Re} s > 1$ the Mellin transforms are*

$$\int_0^\infty \Big(\sum_{n\ge1} \mathfrak{k}(nt)\Big) t^{s-1}\,dt = \zeta(s)\widehat{F}(s), \qquad \int_0^\infty \Big(\sum_{n\ge1} \mu(n)\mathfrak{k}(nt)\Big) t^{s-1}\,dt = \frac{\widehat{F}(s)}{\zeta(s)}. \tag{9.5}$$

Proof. $\mathfrak{k}(nt) \ne 0$ requires $nt \le 1$, i.e. $n \le x$, and then $\mathfrak{k}(nt) = F(1/(nt)) = F(x/n)$. So $\sum_n \mathfrak{k}(nt) = \sum_{n\le x} F(x/n)$, and subtracting $(\log 2 - C)x$ from (9.3) gives $R(x)$; the second sum is (4.1). For (9.5), $\int_0^\infty \mathfrak{k}(nt)t^{s-1}dt = n^{-s}\widehat{F}(s)$ by the substitution $u = nt$, and $\sum_n n^{-\operatorname{Re} s}\int_0^\infty |\mathfrak{k}(u)|u^{\operatorname{Re} s-1}du < \infty$ justifies the interchange by Fubini's theorem. The second transform is (4.7) in the variable $t = 1/x$. □

The first operator in (9.4), $\mathfrak{k} \mapsto \sum_n \mathfrak{k}(n\,\cdot) - (1/t)\int \mathfrak{k}$, is the Müntz (co-Poisson) transform of the strong Nyman–Beurling criterion of Báez-Duarte [1]. By (9.4) and Lemma 6.2(iii), the Müntz transform of $\mathfrak{k}$ at t is $R(1/t) = -E_G'(1/t)/(2\pi)$: the function whose integer singularities stopped the attempts of Section 12 is the Müntz transform of the log-sine kernel. According to the abstract of [1] (an unrefereed arXiv text), its necessity theorem (RH implies that $\mathfrak{k}$ lies in the L^2 closure of the span of the dilations of its Müntz transform) requires $\mathfrak{k}$ continuously differentiable and vanishing at infinity with an integrability condition on $t|\mathfrak{k}'(t)|$, and its sufficiency theorem requires compact support in addition. The kernel (9.1) is not continuously differentiable: it has logarithmic singularities at $t = 1/n$, $n \ge 1$. So that theorem is not applied here, in either direction, and nothing below depends on it.

For $N \ge 1$ and $\sigma > 0$ define the energy and the Gram matrix

$$\begin{aligned} \mathcal{E}_N(\sigma) &= \int_0^\infty \Big|\sum_{d\le N} \mu(d)\,\mathfrak{k}(dt)\Big|^2 t^{2\sigma-1}\,dt, \\ \Pi_\sigma(d,e) &= \int_0^\infty \mathfrak{k}(dt)\mathfrak{k}(et)\,t^{2\sigma-1}\,dt = \int_{\max(d,e)}^\infty F\Big(\frac{x}{d}\Big)F\Big(\frac{x}{e}\Big)\frac{dx}{x^{2\sigma+1}}, \end{aligned} \tag{9.6}$$

the last form by $x = 1/t$, since $\mathfrak{k}(dt)\mathfrak{k}(et) \ne 0$ requires $x \ge \max(d,e)$.

Reviewed by Claude (review 1)

Proposition 9.2 (Spectral form of the energy)**.** *Let $\sigma > 0$ and $N \ge 1$. Then $\mathcal{E}_N(\sigma) < \infty$, and*

$$\mathcal{E}_N(\sigma) = \frac{1}{2\pi}\int_{-\infty}^\infty |\widehat{F}(\sigma+i\tau)|^2 \Big|\sum_{d\le N} \frac{\mu(d)}{d^{\sigma+i\tau}}\Big|^2 d\tau = \sum_{d,e\le N} \mu(d)\mu(e)\,\Pi_\sigma(d,e), \tag{9.7}$$

$$\Pi_\sigma(d,e) = \frac{\Delta_\sigma(\log(e/d))}{(de)^\sigma}, \qquad \Delta_\sigma(u) = \frac{1}{2\pi}\int_{-\infty}^\infty |\widehat{F}(\sigma+i\tau)|^2 e^{i\tau u}\,d\tau, \qquad \Pi_\sigma(d,d) = \frac{\mathcal{N}_\sigma}{d^{2\sigma}}. \tag{9.8}$$

Here $\tau \mapsto \widehat{F}(\sigma+i\tau)$ lies in $L^2(\mathbb{R})$ with $\frac{1}{2\pi}\int |\widehat{F}(\sigma+i\tau)|^2 d\tau = \mathcal{N}_\sigma$, Δ_σ is real, even, continuous, $|\Delta_\sigma| \le \Delta_\sigma(0) = \mathcal{N}_\sigma$, and the matrix $(\Pi_\sigma(d,e))_{d,e\le N}$ is positive semidefinite. Consequently

$$\sum_{d\le N} \mu(d)^2\,\Pi_\sigma(d,d) \le \mathcal{N}_\sigma\,\zeta(2\sigma) \qquad (\sigma > \tfrac12), \tag{9.9}$$

uniformly in N.

Proof. For $d \ge 1$ let $\mathfrak{k}_d(t) = \mathfrak{k}(dt)$ and $g_d(v) = \mathfrak{k}_d(e^v)e^{\sigma v}$, $v \in \mathbb{R}$. By the substitutions $t = e^v$ and $u = dt$, $\int_{\mathbb{R}} |g_d|\,dv = \int_0^\infty |\mathfrak{k}(dt)|t^{\sigma-1}dt = d^{-\sigma}\int_0^\infty |\mathfrak{k}(u)|u^{\sigma-1}du < \infty$ and $\int_{\mathbb{R}} |g_d|^2 dv = \int_0^\infty \mathfrak{k}(dt)^2 t^{2\sigma-1}dt = d^{-2\sigma}\mathcal{N}_\sigma$. So $g_d \in L^1 \cap L^2(\mathbb{R})$, its Fourier transform is $\int g_d(v)e^{-i\tau v}dv = \int_0^\infty \mathfrak{k}(dt)t^{\sigma+i\tau-1}dt = d^{-\sigma-i\tau}\widehat{F}(\sigma+i\tau)$ by (9.3), and Plancherel's theorem [3, Theorem 9.13, p. 186]

applied to $g = \sum_{d\le N}\mu(d)g_d$ gives the first equality in (9.7), with $\mathcal{E}_N(\sigma) = \int |g|^2 dv < \infty$. With $d = 1$ it gives $\frac{1}{2\pi}\int |\widehat{F}(\sigma+i\tau)|^2 d\tau = \mathcal{N}_\sigma$. Expanding the square in (9.6) gives the second equality in (9.7); the matrix is the Gram matrix of the functions $\mathfrak{k}_d t^{\sigma-1/2}$ in $L^2(0,\infty)$, hence positive semidefinite. Plancherel's theorem with polarization, applied to g_d and g_e, gives

$$\Pi_\sigma(d,e) = \int_{\mathbb{R}} g_d g_e\, dv = \frac{1}{2\pi}\int_{-\infty}^{\infty} d^{-\sigma-i\tau}\,\overline{e^{-\sigma-i\tau}}\,|\widehat{F}(\sigma+i\tau)|^2\, d\tau = \frac{1}{(de)^\sigma}\,\Delta_\sigma\Big(\log\frac{e}{d}\Big), \tag{9.10}$$

where $\overline{\widehat{F}(\sigma+i\tau)} = \widehat{F}(\sigma-i\tau)$ because F is real. Since $|\widehat{F}(\sigma+i\cdot)|^2 \in L^1(\mathbb{R})$ is even and non-negative, Δ_σ is real, even, continuous and bounded by $\Delta_\sigma(0) = \mathcal{N}_\sigma$. The diagonal is $\Pi_\sigma(d,d) = \int \mathfrak{k}(dt)^2 t^{2\sigma-1}dt = d^{-2\sigma}\mathcal{N}_\sigma$, and $\sum_{d\le N}\mu(d)^2 d^{-2\sigma} \le \zeta(2\sigma)$ for $\sigma > \frac12$. □

Reviewed by Claude (review 1)

Theorem 9.3 (A sufficient condition for RH)**.** *If for every $\sigma > \frac12$ one has $\liminf_{N\to\infty}\mathcal{E}_N(\sigma) < \infty$, then RH holds. In particular $\sup_N \mathcal{E}_N(\sigma) < \infty$ for every $\sigma > \frac12$ implies RH.*

Proof. Let $1/N \le t \le 1$ with $x = 1/t \notin \mathbb{Z}$. Then $x \le N$, so every $d \le x$ satisfies $d \le N$, and the terms with $x < d \le N$ vanish because $dt > 1$. Hence $\sum_{d\le N}\mu(d)\mathfrak{k}(dt) = H(x)$ by (9.4), and

$$\mathcal{E}_N(\sigma) \ge \int_{1/N}^1 \Big|\sum_{d\le N}\mu(d)\mathfrak{k}(dt)\Big|^2 t^{2\sigma-1}\,dt = \int_1^N |H(x)|^2 x^{-2\sigma-1}\,dx. \tag{9.11}$$

Letting $N\to\infty$ along a sequence realising the $\liminf$, $\int_1^\infty |H(x)|^2x^{-2\sigma-1}dx < \infty$ for every $\sigma > \frac12$, which is condition (b) of Theorem 4.4, and (b) implies RH there. □

The converse of Theorem 9.3 is not proved: by (9.7), it would need a bound for $\int|\widehat{F}(\sigma+i\tau)|^2|\sum_{d\le N}\mu(d)d^{-\sigma-i\tau}|^2d\tau$ uniform in N, and no such bound is established here. By (9.9), the diagonal of the quadratic form (9.7) is bounded uniformly in N for $\sigma > \frac12$ without any hypothesis, and the sign of μ enters only through the off-diagonal part. The statement

$$\sup_{N\ge1}\sum_{\substack{d,e\le N\\ d\ne e}}\frac{\mu(d)\mu(e)}{(de)^\sigma}\,\Delta_\sigma\Big(\log\frac{e}{d}\Big) < \infty \qquad \text{for every } \sigma > \tfrac12 \tag{9.12}$$

is therefore sufficient for RH by Proposition 9.2 and Theorem 9.3. It is not proved, and it is not known to be necessary.

10. A weighted strong Nyman–Beurling criterion for the log-sine kernel

Throughout this section $\frac12 < \sigma < 1$, and $\mathcal{H}_\sigma = L^2((0,\infty), t^{2\sigma-1}dt)$ with norm $\|\cdot\|_\sigma$. For $\phi \in \mathcal{H}_\sigma$ with $\int_0^\infty|\phi(t)|t^{\sigma-1}dt < \infty$ the Mellin transform $\int_0^\infty \phi(t)t^{s-1}dt$ converges absolutely on $\operatorname{Re} s = \sigma$, and Plancherel's theorem [3, Theorem 9.13, p. 186] in the variable $v = \log t$ gives $\|\phi\|_\sigma^2 = \frac{1}{2\pi}\int_{-\infty}^\infty|\int_0^\infty\phi(t)t^{\sigma+i\tau-1}dt|^2d\tau$; this is the fact used in the proof of Proposition 9.2, and it is used below without further comment. The Müntz transform of $\mathfrak{k}$ is written

$$(P\mathfrak{k})(t) = \sum_{n\ge1}\mathfrak{k}(nt) - \frac1t\int_0^\infty\mathfrak{k} = R(1/t)\quad(0<t\le1), \qquad (P\mathfrak{k})(t) = \frac{C-\log 2}{t}\quad(t>1), \tag{10.1}$$

by (9.4) and (9.3); for $t > 1$ the sum is empty and $R(x) = -(\log 2 - C)x$ for $x < 1$, so the two lines agree with $R(1/t)$ for all $t > 0$ with $1/t \notin \mathbb{Z}$.

Reviewed by Claude (review 1)

Lemma 10.1 (Uniform convexity bound)**.** *Let $\frac12 < \sigma_0 < 1$ and $\varepsilon > 0$. Then $|\zeta(\sigma'+i\tau)| \ll_{\sigma_0,\varepsilon} |\tau|^{(1-\sigma_0)/2+\varepsilon}$ uniformly for $\sigma_0 \le \sigma' \le 2$, $|\tau| \ge 2$.*

Proof. Put $\phi(s) = \zeta(s)(s-1)/(s+1)$, holomorphic on $\operatorname{Re} s \ge \sigma_0$. In the half-plane $\operatorname{Re} s \ge \sigma_0$ one has $\zeta(s) = O(|\tau|^{\kappa_0})$ for some κ_0 [5, §5.1, p. 95], so $\phi(s) = O(e^{\varepsilon'|\tau|})$ for every $\varepsilon' > 0$ in the strip $\sigma_0 \le \operatorname{Re} s \le 2$. On $\operatorname{Re} s = 2$, $|\phi(s)| \le \zeta(2)$ because $|s-1| \le |s+1|$. On $\operatorname{Re} s = \sigma_0$, $|\phi(s)| \le |\zeta(\sigma_0+i\tau)| = O(|\tau|^{(1-\sigma_0)/2+\varepsilon})$ by [5, (5.1.4), p. 96] and the definition of $\mu(\sigma_0)$ there.

The Phragmén–Lindelöf theorem for a strip [4, §5.65, pp. 180–181] gives $\phi(\sigma'+i\tau)=O(|\tau|^{\kappa(\sigma')})$ uniformly for $\sigma_0\le\sigma'\le 2$, with κ linear, $\kappa(\sigma_0)=(1-\sigma_0)/2+\varepsilon$, $\kappa(2)=0$, hence $\kappa(\sigma')\le(1-\sigma_0)/2+\varepsilon$. Finally $|\zeta(s)|=|\phi(s)||s+1|/|s-1|\le 2|\phi(s)|$ for $|\tau|\ge 2$, since $|s+1|\le|s-1|+2\le 2|s-1|$. □

Reviewed by Claude (review 1)

Lemma 10.2 (R in the weighted spaces)**.** (i) $\int_1^X R(x)^2dx\le\frac{\pi^2}{12}X^3+2(\log 2-C)^2X^3$ *for* $X\ge 1$, *so that* $\widehat R(s)=\int_1^\infty R(x)x^{-s-1}dx$ *converges absolutely for* $\operatorname{Re}s>1$, *and there*

$$\widehat R(s)=\zeta(s)\widehat F(s)-\frac{\widehat F(1)}{s-1}. \tag{10.2}$$

The right-hand side is holomorphic on $\operatorname{Re}s>0$. (ii) *For every* $\sigma_0>\frac12$, $\int_1^\infty R(x)^2x^{-2\sigma_0-1}dx<\infty$; *the integral in* (10.2) *converges absolutely for* $\operatorname{Re}s>\frac12$ *and* (10.2) *holds there.* (iii) *For* $\frac12<\sigma<1$: $P\mathfrak{k}\in\mathcal{H}_\sigma$, $\int_0^\infty|P\mathfrak{k}(t)|t^{\sigma-1}dt<\infty$, *and* $\int_0^\infty P\mathfrak{k}(t)\,t^{s-1}dt=\zeta(s)\widehat F(s)$ *on* $\operatorname{Re}s=\sigma$.

Proof. (i) By Cauchy–Schwarz over the at most X terms, $(\sum_{\ell\le x}|F(x/\ell)|)^2\le X\sum_{\ell\le x}F(x/\ell)^2$ for $x\le X$; and $\int_\ell^X F(x/\ell)^2dx\le X\pi^2/12$ for each $\ell\le X$ (periodicity, $\int_0^1F^2=\pi^2/12$), so $\int_1^X(\sum_{\ell\le x}|F(x/\ell)|)^2dx\le X\cdot X\cdot X\pi^2/12$, and $(a+b)^2\le 2a^2+2b^2$ with $b=(\log 2-C)x$ gives the bound. Dyadic decomposition and Cauchy–Schwarz give $\int_1^\infty|R|x^{-\operatorname{Re}s-1}dx<\infty$ for $\operatorname{Re}s>1$. For such s, $\sum_\ell\int_\ell^\infty|F(x/\ell)|x^{-\operatorname{Re}s-1}dx=\zeta(\operatorname{Re}s)\int_1^\infty|F(v)|v^{-\operatorname{Re}s-1}dv<\infty$ by the substitution $x=\ell v$, so Fubini's theorem gives $\int_1^\infty\sum_{\ell\le x}F(x/\ell)x^{-s-1}dx=\zeta(s)\widehat F(s)$, and $\int_1^\infty\widehat F(1)x\cdot x^{-s-1}dx=\widehat F(1)/(s-1)$. The residue of $\zeta(s)\widehat F(s)$ at $s=1$ is $\widehat F(1)$, so the right-hand side of (10.2) is holomorphic on $\operatorname{Re}s>0$ ($\widehat F$ is holomorphic on $\operatorname{Re}s>-1$ by Theorem 2.2).

(ii) Fix $\sigma_0\in(\frac12,1)$ and $\varepsilon\in(0,(2\sigma_0-1)/4)$. For $\sigma_0\le\sigma'\le\frac32$ and $|\tau|\ge 2$, (2.9) with $|s|\ge|\tau|$, $|\cot(\pi s/2)|\le\coth(\pi|\tau|/2)\le\coth\pi$ and $|A(s)|\le c\log|\tau|/|\tau|$ (the bound on A obtained in the proof of Theorem 4.4 from $|2k-s|\ge\max(|\tau|,k/2)$) gives $|\widehat F(\sigma'+i\tau)|\le(\pi\coth\pi\,|\zeta(\sigma'+i\tau)|+c\log|\tau|)/|\tau|$, so by Lemma 10.1 and (10.2)

$$|\widehat R(\sigma'+i\tau)|\ll|\tau|^{-\sigma_0+2\varepsilon}+|\tau|^{-(1+\sigma_0)/2+\varepsilon}\log|\tau|+|\tau|^{-1}\ll|\tau|^{-\sigma_0+2\varepsilon}, \tag{10.3}$$

uniformly in σ', and $2\sigma_0-4\varepsilon>1$; on $|\tau|\le 2$ the function is bounded. Hence $\sup_{\sigma_0\le\sigma'\le3/2}\int|\widehat R(\sigma'+i\tau)|^2d\tau<\infty$. For $\sigma'\ge\frac32$ the function $v\mapsto R(e^v)e^{-\sigma' v}$ lies in $L^1\cap L^2(0,\infty)$ by (i), so Plancherel's theorem gives $\int|\widehat R(\sigma'+i\tau)|^2d\tau=2\pi\int_1^\infty R^2x^{-2\sigma'-1}dx\le 2\pi\int_1^\infty R^2x^{-4}dx$. Thus $\sup_{\sigma'>\sigma_0}\int|\widehat R(\sigma'+i\tau)|^2d\tau<\infty$. Put $\mathcal{P}(z)=\widehat R(\sigma_0-iz)$ for $\operatorname{Im}z>0$. By the Paley–Wiener theorem [3, Theorem 19.2, pp. 372–373] there is $\Psi\in L^2(0,\infty)$ with $\mathcal{P}(z)=\int_0^\infty\Psi(v)e^{ivz}dv$; for $\upsilon>\frac32-\sigma_0$ also $\mathcal{P}(\xi+i\upsilon)=\int_0^\infty R(e^v)e^{-(\sigma_0+\upsilon)v}e^{i\xi v}dv$, and both $\Psi(v)e^{-\upsilon v}$ and $R(e^v)e^{-(\sigma_0+\upsilon)v}$ lie in $L^1(0,\infty)$ with the same Fourier transform, so they agree almost everywhere [3, Theorem 9.12, p. 185]. Hence $\int_1^\infty R^2x^{-2\sigma_0-1}dx=\int_0^\infty|\Psi|^2<\infty$. Absolute convergence of $\widehat R(s)$ for $\operatorname{Re}s>\frac12$ follows by Cauchy–Schwarz, $\int_1^\infty|R|x^{-\operatorname{Re}s-1}\le(\int R^2x^{-2\sigma_0-1})^{1/2}(\int x^{2\sigma_0-2\operatorname{Re}s-1})^{1/2}$ with $\frac12<\sigma_0<\operatorname{Re}s$, locally uniformly, so the integral is holomorphic on $\operatorname{Re}s>\frac12$ and agrees with the right-hand side of (10.2) there by the identity theorem.

(iii) By (10.1) and $x=1/t$: $\int_0^1|P\mathfrak{k}|^2t^{2\sigma-1}dt=\int_1^\infty R^2x^{-2\sigma-1}dx<\infty$ by (ii), and $\int_1^\infty(\log 2-C)^2t^{2\sigma-3}dt=(\log 2-C)^2/(2-2\sigma)$ since $\sigma<1$. Likewise $\int_0^1|P\mathfrak{k}|t^{\sigma-1}dt=\int_1^\infty|R|x^{-\sigma-1}dx<\infty$ by (ii), and $\int_1^\infty t^{\sigma-2}dt<\infty$. On $\operatorname{Re}s=\sigma$, $\int_0^1P\mathfrak{k}(t)t^{s-1}dt=\widehat R(s)$ and $\int_1^\infty(C-\log 2)t^{s-2}dt=(\log 2-C)/(s-1)=\widehat F(1)/(s-1)$, and (10.2) gives the sum $\zeta(s)\widehat F(s)$. □

For $n\ge1$ let

$$\varrho_n(t)=(P\mathfrak{k})(nt)-\frac1n(P\mathfrak{k})(t)\qquad(t>0). \tag{10.4}$$

For $t>1$ both terms equal $(C-\log 2)/(nt)$, so $\varrho_n(t)=0$: ϱ_n is supported in $(0,1]$. By Lemma 10.2(iii) and the substitution $u=nt$, $\varrho_n\in\mathcal{H}_\sigma$, $\int_0^\infty|\varrho_n|t^{\sigma-1}dt<\infty$, and its Mellin transform on $\operatorname{Re}s=\sigma$ is $(n^{-s}-n^{-1})\zeta(s)\widehat F(s)$. For a Dirichlet polynomial $D(s)=\sum_{n\le N}c_nn^{-s}$ with

real c_n put $\tilde D(s) = D(s) - D(1)$, so that $\tilde D(1) = 0$; then

$$\int_0^\infty \Big(\mathfrak{k}(t) - \sum_{n\le N} c_n \varrho_n(t)\Big) t^{s-1} dt = \widehat F(s)(1 - \zeta(s)\tilde D(s)) \qquad (\operatorname{Re} s = \sigma), \tag{10.5}$$

and the function $\Phi_D(s) = \widehat F(s)(1-\zeta(s)\tilde D(s))$ is holomorphic on $\operatorname{Re} s > 0$, the pole of ζ at $s=1$ being cancelled by $\tilde D(1) = 0$. Conversely every Dirichlet polynomial E with $E(1) = 0$ is $\tilde D$ for $D = E$.

Reviewed by Claude (review 1)

Theorem 10.3 (Weighted strong criterion)**.** *Let $\frac12 < \sigma < 1$.* (i) *If $\mathfrak{k}$ lies in the $\mathcal{H}_\sigma$-closure of the linear span of $\{\varrho_n : n \ge 1\}$, then $\zeta(s) \ne 0$ for $\operatorname{Re} s > \sigma$.* (ii) *Assume RH. Then for every $\varepsilon > 0$,*

$$\Big\| \mathfrak{k} - \sum_{n\le N} \mu(n)\varrho_n \Big\|_\sigma \ll_{\sigma,\varepsilon} N^{-(\sigma-1/2)+\varepsilon} \qquad (N \ge 2). \tag{10.6}$$

Consequently RH holds if and only if $\mathfrak{k}$ lies in the $\mathcal{H}_\sigma$-closure of $\operatorname{span}\{\varrho_n\}$ *for every $\sigma \in (\frac12, 1)$.*

Proof. (i) Suppose $\zeta(\rho) = 0$ with $\operatorname{Re}\rho > \sigma$; then $\operatorname{Re}\rho < 1$. Let $\eta > 0$ and choose real c_n with $\|\phi\|_\sigma < \eta$, where $\phi = \mathfrak{k} - \sum_{n\le N} c_n\varrho_n$. The function ϕ is supported in $(0,1]$, so for $\sigma' \ge \sigma$, $\|\phi\|_{\sigma'}^2 = \int_0^1 |\phi|^2 t^{2\sigma'-1}dt \le \|\phi\|_\sigma^2 < \eta^2$, and $\int_0^1 |\phi| t^{\sigma'-1}dt \le \int_0^1 |\phi| t^{\sigma-1}dt < \infty$. For $\operatorname{Re} s > \sigma$ the Mellin integral of ϕ therefore converges absolutely, is holomorphic, and equals $\Phi_D(s)$: the computation that gives (10.5) holds on every line $\operatorname{Re} s = \sigma' \in [\sigma, 1)$ (Lemma 10.2(iii) applies to every such σ'), so the two holomorphic functions agree on the open strip $\sigma < \operatorname{Re} s < 1$, hence on $\operatorname{Re} s > \sigma$ by the identity theorem. Plancherel on each line gives $\frac{1}{2\pi}\int |\Phi_D(\sigma' + i\tau)|^2 d\tau = \|\phi\|_{\sigma'}^2 < \eta^2$ for all $\sigma' \ge \sigma$. Put $\mathcal{P}(z) = \Phi_D(\sigma - iz)$ for $\operatorname{Im} z > 0$. By the Paley–Wiener theorem [3, Theorem 19.2, pp. 372–373] there is $\Psi \in L^2(0,\infty)$ with $\mathcal{P}(z) = \int_0^\infty \Psi(v) e^{ivz} dv$ and $\int_0^\infty |\Psi|^2 = \sup_{\upsilon>0} \frac{1}{2\pi}\int |\mathcal{P}(\xi + i\upsilon)|^2 d\xi \le \eta^2$. For $z = \xi + i\upsilon$, Cauchy–Schwarz gives $|\mathcal{P}(z)|^2 \le \int |\Psi|^2 \int_0^\infty e^{-2\upsilon v} dv \le \eta^2/(2\upsilon)$. With z such that $\sigma - iz = \rho$, i.e. $\upsilon = \operatorname{Re}\rho - \sigma > 0$,

$$|\Phi_D(\rho)|^2 \le \frac{\eta^2}{2(\operatorname{Re}\rho - \sigma)}. \tag{10.7}$$

But $\Phi_D(\rho) = \widehat F(\rho)(1 - \zeta(\rho)\tilde D(\rho)) = \widehat F(\rho) = A(\rho)$ by (2.9), and $A(\rho) \ne 0$ by Theorem 3.1. Since $\eta > 0$ is arbitrary, this is a contradiction.

(ii) Assume RH. Fix δ with $0 < \delta < \sigma - \frac12$, put $\eta_0 = \sigma - \frac12 - \delta > 0$, and let $\varepsilon \in (0, \eta_0/8)$. Under RH, for $\frac12 + \delta \le \operatorname{Re} u \le 2$ and $|\operatorname{Im} u| \ge 2$,

$$|\zeta(u)| \ll_{\delta,\varepsilon} |\operatorname{Im} u|^\varepsilon, \qquad |1/\zeta(u)| \ll_{\delta,\varepsilon} |\operatorname{Im} u|^\varepsilon : \tag{10.8}$$

for $\frac12 + \delta \le \operatorname{Re} u \le 1$ this is [5, Theorem 14.2, (14.2.1), p. 336], which is uniform in that range and gives $|\log\zeta(u)| \ll (\log|\operatorname{Im} u|)^{1-2\delta+\varepsilon}$; for $1 \le \operatorname{Re} u \le 2$, $|\zeta(u)| \ll \log|\operatorname{Im} u|$ by [5, Theorem 4.11, p. 77] with a partial sum of length $|\operatorname{Im} u|$, and $|1/\zeta(u)| \ll \log^7|\operatorname{Im} u|$ by [5, (3.6.5), p. 51]. On the compact set $\frac12 + \delta \le \operatorname{Re} u \le 2$, $|\operatorname{Im} u| \le 2$, $1/\zeta$ is continuous under RH (no zeros, and $1/\zeta(1) = 0$), hence bounded. So $|1/\zeta(u)| \ll (2 + |\operatorname{Im} u|)^\varepsilon$ on the whole strip.

Let $N \ge 2$, $x = N + \frac12$, $T = N^2$, $c = 2 - \sigma$, and $s = \sigma + i\tau$. The lemma of [5, §3.12, Lemma 3.12, pp. 60–61] applied to $f = 1/\zeta$ (coefficients $\mu(n)$, so $\psi \equiv 1$ and $\sum |\mu(n)| n^{-\sigma'} \le \zeta(\sigma') \ll (\sigma'-1)^{-1}$, i.e. $\alpha = 1$), with $\sigma + c = 2 > 1$ and $|x - N| = \frac12$, gives

$$D_N(s) := \sum_{n\le N} \frac{\mu(n)}{n^s} = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} \frac{x^w}{\zeta(s+w)} \frac{dw}{w} + O\Big(\frac{x^c}{T}\Big) + O\Big(\frac{x^{1-\sigma}\log x}{T}\Big), \tag{10.9}$$

and the error terms are $\ll N^{-\sigma}$. Move the segment to $\operatorname{Re} w = -\eta_0$, so that $\operatorname{Re}(s+w) = \frac12 + \delta$. In the rectangle $-\eta_0 \le \operatorname{Re} w \le c$, $|\operatorname{Im} w| \le T$, the integrand has the single pole $w = 0$, with residue $1/\zeta(s)$: $1/\zeta(s+w)$ is holomorphic there under RH, and $w = 1 - s$ is a zero of it. On the horizontal

segments $w = v \pm iT$, $-\eta_0 \le v \le c$, by (10.8) $|1/\zeta(s+w)| \ll (T+|\tau|+2)^{\varepsilon}$ and $|x^w/w| \le x^c/T$, so their contribution is $\ll (N^2+|\tau|)^{\varepsilon} N^{-\sigma}$. On the vertical segment $w = -\eta_0 + iv$, $|v| \le T$,

$$\Big|\int \frac{x^w}{\zeta(s+w)}\frac{dw}{w}\Big| \ll x^{-\eta_0}\int_{-T}^{T}\frac{(2+|\tau|+|v|)^{\varepsilon}}{\eta_0+|v|}\,dv \ll_{\eta_0} N^{-\eta_0}(N^2+|\tau|)^{\varepsilon}\log N. \tag{10.10}$$

Since $(N^2+|\tau|)^{\varepsilon} \le N^{2\varepsilon}(2+|\tau|)^{\varepsilon}$ and $N^{-\sigma} \le N^{-\eta_0}$,

$$\Big|D_N(s) - \frac{1}{\zeta(s)}\Big| \ll_{\sigma,\delta,\varepsilon} N^{-\eta_0+3\varepsilon}(2+|\tau|)^{\varepsilon} \qquad (\operatorname{Re} s = \sigma). \tag{10.11}$$

Multiplying by $\zeta(s)$ and using $|\zeta(\sigma+i\tau)| \ll (2+|\tau|)^{\varepsilon}$ (by (10.8) for $|\tau| \ge 2$, and boundedness for $|\tau| \le 2$, $\sigma < 1$),

$$|1-\zeta(s)D_N(s)| \ll N^{-\eta_0+3\varepsilon}(2+|\tau|)^{2\varepsilon} \qquad (\operatorname{Re} s = \sigma). \tag{10.12}$$

Now $\tilde{D}_N = D_N - m(N)$, and by (10.5), Plancherel and (10.12),

$$\Big\|\mathfrak{k} - \sum_{n\le N}\mu(n)\varrho_n\Big\|_{\sigma} \le \Big(\frac{1}{2\pi}\int |\widehat{F}(\sigma+i\tau)|^2|1-\zeta D_N|^2 d\tau\Big)^{1/2} + |m(N)|\,\|P\mathfrak{k}\|_{\sigma}, \tag{10.13}$$

where the second term comes from $\widehat{F}(s)\zeta(s)m(N)$, the Mellin transform of $m(N)P\mathfrak{k}$ by Lemma 10.2(iii). The first term is at most $N^{-\eta_0+3\varepsilon}$ times a constant multiple of the square root of $\int |\widehat{F}(\sigma+i\tau)|^2(2+|\tau|)^{4\varepsilon}d\tau$, and this integral is finite because $|\widehat{F}(\sigma+i\tau)| \ll (|\zeta(\sigma+i\tau)|+\log|\tau|)/|\tau| \ll |\tau|^{-1+\varepsilon}$ for $|\tau| \ge 2$ under RH, with $6\varepsilon < 1$. The second term is $\ll N^{-1/2+\varepsilon}$ by (6.15) and Lemma 10.2(iii). Altogether the norm is $\ll N^{-\eta_0+3\varepsilon} + N^{-1/2+\varepsilon} \ll N^{-(\sigma-1/2)+\delta+3\varepsilon}$, since $\eta_0 = \sigma - \frac12 - \delta$ and $\frac12 \ge \sigma - \frac12$. Given $\varepsilon' > 0$, choosing $\delta = \varepsilon'/2$ and $\varepsilon = \varepsilon'/6$ gives the exponent $-(\sigma-\frac12)+\varepsilon'$ of (10.6).

The equivalence: (ii) gives one direction; for the other, (i) for every $\sigma \in (\frac12, 1)$ gives $\zeta(s) \ne 0$ for $\operatorname{Re} s > \frac12$, which is RH. □

The classical strong Nyman–Beurling criterion is the case of the kernel $\mathbf{1}_{(0,1]}$, whose Müntz transform is $-\{1/t\}$ and whose Mellin transform is $1/s$; Theorem 10.3 replaces $1/s$ by $\widehat{F}$, the weight $1/|s|^2$ by $|\widehat{F}(\sigma+i\tau)|^2$, and the space $L^2(0,\infty)$ by $\mathcal{H}_\sigma$. The two ingredients that make the kernel $\mathfrak{k}$ admissible, although it is not continuously differentiable, are the support of $\mathfrak{k} - \sum c_n\varrho_n$ in $(0,1]$, which makes the weighted norms monotone in σ', and $A \ne 0$ at the zeros, which is what (10.7) contradicts.

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Proposition 10.4 (Finite prime cut)**.** *For $Y \ge 2$ let $\mathcal{S}_Y$ be the set of squarefree integers all of whose prime factors are $\le Y$, and*

$$\Pi_Y(s) = \prod_{p\le Y}(1-p^{-s}) = \sum_{n\in\mathcal{S}_Y}\mu(n)n^{-s}, \qquad \zeta_{>Y}(s) = \zeta(s)\Pi_Y(s), \tag{10.14}$$

so that $\zeta_{>Y}(s) = \prod_{p>Y}(1-p^{-s})^{-1}$ for $\operatorname{Re} s > 1$, and $\zeta_{>Y}$ has the same zeros as ζ in $0 < \operatorname{Re} s < 1$. Let $\frac12 < \sigma < 1$. If

$$\liminf_{Y\to\infty}\int_{-\infty}^{\infty}|\widehat{F}(\sigma+i\tau)|^2\,|1-\zeta_{>Y}(\sigma+i\tau)|^2\,d\tau = 0, \tag{10.15}$$

then $\zeta(s) \ne 0$ for $\operatorname{Re} s > \sigma$. In particular, (10.15) for every $\sigma \in (\frac12, 1)$ implies RH.

Proof. $\mathcal{S}_Y$ is finite, so Π_Y is a Dirichlet polynomial, and $\Pi_Y(1) = \prod_{p\le Y}(1-1/p) \le (\sum_{n\le Y}1/n)^{-1} \le 1/\log Y$, because $\prod_{p\le Y}(1-1/p)^{-1} = \prod_{p\le Y}\sum_{k\ge0}p^{-k} \ge \sum_{n\le Y}1/n$. By (10.5) with $D = \Pi_Y$ and Plancherel,

$$\Big\|\mathfrak{k} - \sum_{n\in\mathcal{S}_Y}\mu(n)\varrho_n\Big\|_{\sigma} \le \Big(\frac{1}{2\pi}\int |\widehat{F}(\sigma+i\tau)|^2|1-\zeta_{>Y}(\sigma+i\tau)|^2 d\tau\Big)^{1/2} + \frac{\|P\mathfrak{k}\|_{\sigma}}{\log Y}, \tag{10.16}$$

which tends to 0 along a sequence of Y by (10.15) and Lemma 10.2(iii). Theorem 10.3(i) applies. □

The converse of Proposition 10.4, that RH implies (10.15), is not proved here. The proof of Theorem 10.3(ii) does not transfer: it rests on Perron's formula for the sharp cutoff $n \le N$, whose error terms are controlled by the length N, whereas Π_Y has length $\prod_{p\le Y} p$, and no bound for $|\zeta_{>Y}(\sigma+i\tau)|$ uniform in Y on the line $\operatorname{Re} s = \sigma < 1$ is available to us. The factor $\Pi_Y(s)$ is the inverse of the Euler product of ζ truncated at Y; as $Y \to \infty$ it converges to $1/\zeta(s)$ for $\operatorname{Re} s > 1$ only, and on the line $\operatorname{Re} s = \sigma < 1$ the statement (10.15) is a statement about ζ itself, not about a function with finitely many parameters.

11. The corner as a weighted Liouville sum, and the Clausen kernel

Write $U = \sqrt{X}$ and group the corner by $n = dj$:

$$\mathcal{C}(X) = \sum_{U<n\le X} a_U(n)\, n\, E_G\Big(\frac{X}{n}\Big), \qquad a_U(n) = \sum_{\substack{dj=n \\ d\le U,\ j\le U}} \mu(d)\mu(j). \tag{11.1}$$

Let $\lambda(n) = (-1)^{\Omega(n)}$ be Liouville's function, $\Omega(n)$ the number of prime factors of n counted with multiplicity.

Reviewed by Claude (review 1)

Proposition 11.1 (Liouville form of the corner)**.** *For $n \ge 1$ and $U \ge 1$, $a_U(n) = \lambda(n)\, b_U(n)$ with $b_U(n) \ge 0$ the number of factorizations $n = dj$ with $d, j \le U$ both squarefree. Explicitly, $b_U(n) = 0$ unless $n = ab^2$ with a, b squarefree and $(a, b) = 1$, and then*

$$b_U(n) = \#\Big\{t \mid a :\ \frac{ab}{U} \le t \le \frac{U}{b}\Big\}. \tag{11.2}$$

Hence

$$\mathcal{C}(X) = \sum_{\sqrt{X}<n\le X} \lambda(n)\, b_{\sqrt{X}}(n)\, n\, E_G\Big(\frac{X}{n}\Big), \tag{11.3}$$

and by Theorem 6.4, RH is equivalent to $\sum_{\sqrt{X}<n\le X} \lambda(n) b_{\sqrt{X}}(n)\, nE_G(X/n) \ge -K_\varepsilon X^{3/2+\varepsilon}$ for all $X \ge 1$, for every $\varepsilon > 0$.

Proof. If $\mu(d)\mu(j) \ne 0$ then d and j are squarefree, so $\Omega(d) = \omega(d)$, $\Omega(j) = \omega(j)$, and $\mu(d)\mu(j) = (-1)^{\omega(d)+\omega(j)} = (-1)^{\Omega(dj)} = \lambda(n)$. Thus every nonzero term of $a_U(n)$ equals $\lambda(n)$, which gives $a_U = \lambda b_U$ with $b_U(n)$ the number of such pairs. If $n = dj$ with d, j squarefree, put $b = (d, j)$ and $a = n/b^2 = (d/b)(j/b)$; the factors d/b and j/b are coprime and squarefree, so a is squarefree, and a prime dividing b divides neither d/b nor j/b (as d, j are squarefree), so $(a, b) = 1$. Conversely, given $n = ab^2$ with a, b squarefree and coprime, the pairs are exactly $d = bt$, $j = ab/t$ with $t \mid a$ (both squarefree because $(t, b) = 1$), and the condition $d, j \le U$ is $ab/U \le t \le U/b$. This is (11.2); if n has a cube factor it is not of the form ab^2 with a, b squarefree, so $b_U(n) = 0$. The identity (11.2) was also checked against the definition for every $n \le U^2$ with $U = 7, 20, 45$ (`liouville_corner.py`). □

In (11.3) the sign of each term is $\lambda(n) \operatorname{sgn} E_G(X/n)$ and the weight $b_{\sqrt{X}}(n)$ is nonnegative: there is no cancellation inside $a_U(n)$, only between different n. The weight counts the divisors t of a in the multiplicative window $[ab/U, U/b]$, whose ratio is $U^2/(ab^2) = X/n$; it vanishes unless the largest prime factor of n is at most $\sqrt{X}$.

Reviewed by Claude (review 1)

Proposition 11.2 (The Clausen kernel and its Müntz transform)**.** *Let $\mathfrak{c}(t) = t\,\mathrm{Cl}_2(2\pi/t)$ for $0 < t \le 1$ and $\mathfrak{c}(t) = 0$ for $t > 1$. Then $\int_0^\infty \mathfrak{c}(t) t^{s-1} dt = \Theta(s)$ for $\operatorname{Re} s > 0$, $\int_0^\infty \mathfrak{c} = a$, and for $Y \ge 1$, $Y \notin \mathbb{Z}$,*

$$\frac{E_G(Y)}{Y} = \sum_{\ell\ge 1} \mathfrak{c}\Big(\frac{\ell}{Y}\Big) - Y\int_0^\infty \mathfrak{c} = (P\mathfrak{c})(1/Y), \tag{11.4}$$

the Müntz transform of $\mathfrak{c}$ *at* $t = 1/Y$. *Consequently*

$$\mathcal{C}(X) = X \sum_{\sqrt{X}<n\le X} \lambda(n)\, b_{\sqrt{X}}(n)\, (P\mathfrak{c})\Big(\frac{n}{X}\Big), \tag{11.5}$$

and, with $D_U(s) = \sum_{n\le U} \mu(n) n^{-s}$ *and*

$$L_U(s) = \sum_{U<n\le X} \lambda(n) b_U(n) n^{-s} = D_U(s)^2 - \sum_{n\le U} (\mu*\mu)(n) n^{-s}, \tag{11.6}$$

$$\mathcal{C}(X) = \frac{1}{2\pi i}\int_{(5/2)} \zeta(s-1)\Theta(s)\, L_U(s-1)\, X^s\, ds - aX^2 \sum_{U<n\le X} \frac{\lambda(n) b_U(n)}{n}. \tag{11.7}$$

Proof. With $v = 1/t$, $\int_0^1 t\,\mathrm{Cl}_2(2\pi/t) t^{s-1} dt = \int_1^\infty \mathrm{Cl}_2(2\pi v) v^{-s-1} dv = \Theta(s)$ by (6.3), and $\Theta(2) = a$. Next $\mathfrak{c}(\ell/Y) = (\ell/Y)\mathrm{Cl}_2(2\pi Y/\ell)$ for $\ell \le Y$ and 0 for $\ell > Y$, so $\sum_\ell \mathfrak{c}(\ell/Y) = G(Y)/Y$, and subtracting $aY = Y\int \mathfrak{c}$ gives $E_G(Y)/Y$; this is the Müntz transform $\sum_\ell \mathfrak{c}(\ell t) - t^{-1}\int \mathfrak{c}$ at $t = 1/Y$. Then (11.5) is (11.3) with $nE_G(X/n) = X \cdot E_G(X/n)/(X/n)$. For (11.7), (6.6) gives $G(X/n) = \frac{1}{2\pi i}\int_{(5/2)} \zeta(s-1)\Theta(s)(X/n)^s ds$ for each of the finitely many n, and $E_G = G - aY^2$; the series $\sum_{d,j\le U} \mu(d)\mu(j)(dj)^{-s} = D_U(s)^2$ splits into $dj \le U$ and $dj > U$, which gives the formula for L_U. □

Thus the corner is X times a weighted Liouville sum of dilations of the Müntz transform of the Clausen kernel $\mathfrak{c}$, the kernel one integration above $\mathfrak{k}$ ($\mathfrak{k}(t) = F(1/t)$, $\mathfrak{c}(t) = t\,\mathrm{Cl}_2(2\pi/t)$, and $\mathrm{Cl}_2' = -F(\cdot/2\pi)$ by (2.1)). The Mellin factor $\zeta(s-1)\Theta(s)$ in (11.7) is the transform of G; the arithmetic is entirely in $L_U(s-1) = D_U(s-1)^2 - \sum_{n\le U}(\mu*\mu)(n) n^{1-s}$, a Dirichlet polynomial of length $U^2 = X$ built from μ on $[1, \sqrt{X}]$ only.

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Proposition 11.3 (One-sided Liouville–Clausen criterion)**.** *Let* $\Lambda_G(X) = \sum_{n\le X} \lambda(n)\, n\, E_G(X/n)$ *for* $X \ge 1$. *Suppose that for every* $\varepsilon > 0$ *there is* K_ε *such that either* $\Lambda_G(X) \le K_\varepsilon X^{3/2+\varepsilon}$ *for all* $X \ge 1$, *or* $\Lambda_G(X) \ge -K_\varepsilon X^{3/2+\varepsilon}$ *for all* $X \ge 1$. *Then RH holds.*

Proof. By Lemma 6.2(ii) with $\delta = \frac12$,

$$|\Lambda_G(X)| \le \sum_{n\le X} n\, c_{1/2}\Big(\frac{X}{n}\Big)^{3/2} \le c_{1/2} X^{3/2} \sum_{n\le X} n^{-1/2} \le 2c_{1/2} X^2, \tag{11.8}$$

so $\widehat{\Lambda}_G(s) = \int_1^\infty \Lambda_G(X) X^{-s-1} dX$ converges absolutely for $\mathrm{Re}\, s > 2$. There, by Fubini (the same bound gives absolute convergence of the double integral), $\widehat{\Lambda}_G(s) = \sum_n \lambda(n) n^{1-s} \int_n^\infty E_G(X/n) X^{-s-1} dX = \widehat{E}_G(s) \sum_n \lambda(n) n^{1-s}$, where $\widehat{E}_G(s) = \int_1^\infty E_G Y^{-s-1} dY = \zeta(s-1)\Theta(s) - a/(s-2)$ by Lemma 6.2(i), and $\sum_n \lambda(n) n^{-w} = \zeta(2w)/\zeta(w)$ for $\mathrm{Re}\, w > 1$. Hence

$$\widehat{\Lambda}_G(s) = \Big(\zeta(s-1)\Theta(s) - \frac{a}{s-2}\Big)\frac{\zeta(2s-2)}{\zeta(s-1)} \qquad (\mathrm{Re}\, s > 2). \tag{11.9}$$

The right-hand side is meromorphic on $\mathrm{Re}\, s > 1$; on the real half-line $s > \frac32$ it is holomorphic, since $\zeta(s-1) \ne 0$ for real $s - 1 > 0$ (proof of Proposition 5.1(iii)), the first factor is holomorphic on $\mathrm{Re}\, s > 0$, and the only real pole of $\zeta(2s-2)$ is $s = \frac32$. Fix $\varepsilon > 0$, $\beta = \frac32 + \varepsilon$, and suppose $\Lambda_G(X) \le K_\varepsilon X^\beta$. Then $f = K_\varepsilon X^\beta - \Lambda_G$ is nonnegative, locally bounded and measurable, its Mellin integral $g(s) = K_\varepsilon/(s-\beta) - \widehat{\Lambda}_G(s)$ converges absolutely for $\mathrm{Re}\, s > 2$, and g extends holomorphically to a neighbourhood of every real segment $[\sigma_2, 2]$ with $\sigma_2 > \beta$; Lemma 5.2 gives absolute convergence of the Mellin integral of f, hence of $\widehat{\Lambda}_G$, on $\mathrm{Re}\, s > \beta$, so the right-hand side of (11.9) is holomorphic there. Let ρ be a zero of ζ with $\frac12 + \varepsilon < \mathrm{Re}\,\rho < 1$. At $s = 1+\rho$ the denominator vanishes, $\zeta(2s-2) = \zeta(2\rho) \ne 0$ because $\mathrm{Re}\, 2\rho > 1$, and the first factor equals $-a/(\rho-1) \ne 0$ because $\zeta(\rho)\Theta(1+\rho) = 0$. So $s = 1+\rho$ would be a pole in $\mathrm{Re}\, s > \beta$, a contradiction.

The other one-sided bound is handled with $f = \Lambda_G + K_\varepsilon X^\beta$. Letting $\varepsilon \to 0$ and using $\zeta \neq 0$ on $\operatorname{Re} s \geq 1$ gives RH. □

The converse, RH $\Rightarrow \Lambda_G(X) \ll X^{3/2+\varepsilon}$, is not written out here. Proposition 11.3 uses λ on all of $[1, X]$ and no weight; it is recorded because the measurements below show that the weight $b_{\sqrt{X}}$ in (11.3) does not change the scale of the sum.

Reviewed by Claude (review 1)

Remark 11.4 (Measurements on the Liouville form). At $X = 30000.5$, 120000.5, 380000.5 (`liouville_weight.py`): $b_{\sqrt{X}}(n) > 0$ for 14.2%, 13.9%, 13.8% of the $n \in (\sqrt{X}, X]$, with mean 2.48, 2.61, 2.66 on its support and maximum 16, 22, 26. The contributions of the n with a fixed value $b_{\sqrt{X}}(n) = k$ are each of size 0.01–$0.2\,X^{3/2}$ with varying signs ($k = 2$: -0.089, -0.085, -0.047; $k = 4$: $+0.220$, -0.077, $+0.189$). The indicator sum $\sum_{b>0} \lambda(n) n E_G(X/n)$ is $+0.026$, -0.013, $+0.033\,X^{3/2}$, and the unweighted $\sum_{\sqrt{X}<n\leq X} \lambda(n) n E_G(X/n)$ is -0.059, -0.095, $+0.011\,X^{3/2}$, against $\sum_n n|E_G(X/n)| = 81$, 163, $290\,X^{3/2}$.

One-sided Abel form (`corner_onesided.py`). With $g(n) = nE_G(X/n)$, $b_X(n) = g(n+1) - g(n)$ and $A_U(t) = \sum_{U<n\leq t} a_U(n)$, $\mathcal{C}(X) = A_U(\lfloor X \rfloor) g(\lfloor X \rfloor) - \sum_{U<n<\lfloor X \rfloor} A_U(n) b_X(n)$ exactly (residual $< 10^{-6}$), and $\mathcal{C} \geq -K_\varepsilon X^{3/2+\varepsilon}$ follows from $\sum_n [A_U(n) b_X(n)]_+ \ll X^{3/2+\varepsilon}$ plus the boundary term. Measured, in units of $X^{3/2}$: $\sum [A_U b_X]_+ = 3.77$, 7.01, 5.92; $\sum |A_U b_X| = 7.69$, 14.10, 11.95; boundary term -0.007, $+0.009$, $+0.002$; $\max_n |A_U(n)|/\sqrt{X} = 0.36$, 0.45, 0.41. The Abel regrouping brings the sum of absolute values from 26–$95\,X^{3/2}$ (termwise, Table 1) to 8–$14\,X^{3/2}$, no longer growing like $X^{1/2}$ in this range; the positive part is one half of it, so the one-sided form gains nothing over the absolute one here.

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Remark 11.5 (The μ^2-corner). Let $\mathcal{C}_{\mu^2}(X)$ be (6.9) with $\mu(d)\mu(j)$ replaced by $\mu(d)^2\mu(j)^2$. Replacing the sum over squarefree $d = Us$, $j = Ut$ by the integral with density $6/\pi^2$ in each variable suggests $\mathcal{C}_{\mu^2}(X) \sim \kappa X^2$ with

$$\begin{aligned} \kappa &= \Big(\frac{6}{\pi^2}\Big)^2 \int_0^1\!\!\int_0^1 st\, E_G\Big(\frac{1}{st}\Big)\, ds\, dt = \Big(\frac{6}{\pi^2}\Big)^2 \int_1^\infty \frac{E_G(y)\log y}{y^3}\, dy, \\ \int_1^\infty \frac{E_G(y)\log y}{y^3}\, dy &= -\widehat{E}_G{}'(2) = -\frac{\Theta''(2)}{2} - \gamma_E \Theta'(2) + \gamma_1 a, \end{aligned} \tag{11.10}$$

where γ_E is Euler's constant and γ_1 the first Stieltjes constant; the second equality uses $\int_0^1\!\int_0^1 \phi(st)\, ds\, dt = \int_0^1 \phi(p)\log(1/p)\, dp$ and $p = 1/y$, the third $\widehat{E}_G(s) = \zeta(s-1)\Theta(s) - a/(s-2)$ expanded at $s = 2$. Numerically (`musq_corner_const.py`) $\Theta'(2) = -0.0146808536$, $\Theta''(2) = -0.000882717$, the integral is -0.00267067 (confirmed by direct integration of E_G up to $y = 3000$: -0.002671), and $\kappa = -0.000987014$. The passage from the sum to the integral is not proved here, and the measured values of $\mathcal{C}_{\mu^2}(X)/X^2$ at $X = 30000.5$, 120000.5, 380000.5 are -0.000762, -0.000988, -0.001327: they cross κ and drift by about -0.00022 per unit of $\log X$, so the asymptotic κX^2 is not confirmed in this range. What is used in Remark 8.2 is only that $\mathcal{C}_{\mu^2}$ is measured to be negative and of order X^2. The near-diagonal part $|d - j| \leq H$ of the corner has at most $(H+1)\sqrt{X}$ pairs, each $\ll_\delta X^{1+\delta}(dj)^{-\delta}$ by Lemma 6.2(ii), hence is $\ll_\delta (H+1)X^{3/2}$ for $H \leq X^{1/4}/2$; this is a count of pairs, not a cancellation, and it removes nothing from the problem.

12. Attempts on the corner and where they stopped

This section records the methods tried on (14.1), with the measured quantity at which each stopped. None of them is claimed to be impossible; they were not carried through.

Termwise bounds. Lemma 6.2(ii) gives $|\mathcal{C}(X)| \leq \sum_{d,j} dj |E_G(X/(dj))|$, which loses all cancellation from $\mu(d)\mu(j)$. With the measured size $|E_G(Y)| \approx Y^{3/4}$ the sum of absolute values of the terms of $\mathcal{C}$ is 26.0, 53.5, 95.5 in units of $X^{3/2}$ at $X = 3 \cdot 10^4$, $1.2 \cdot 10^5$, $3.8 \cdot 10^5$, while $|\mathcal{C}|$ itself is 0.13, 0.08,

0.11 in the same units (Table 1). In units of $X^{3/2}$ the sum of absolute values grows like $X^{1/2}$: from $X = 3 \cdot 10^4$ to $1.2 \cdot 10^5$ it doubles, and from $1.2 \cdot 10^5$ to $3.8 \cdot 10^5$ it grows by $1.78 \approx (3.8/1.2)^{1/2}$.

Induction through $E'_G = -2\pi R$. Summing the singular parts of R at the integers (Remark 6.5) over the pairs (d, j) was attempted. In the attempts made, the sum over the corner did not show cancellation, and no input on the signs $\mu(d)\mu(j)$ was found that controls it.

Quadratic certificates. By Remark 8.2, a positive semidefinite certificate that uses only the relation $\mu(d)\mu(d') = \mu(dd'/(d,d')^2)$ applies equally to μ^2; for W_{μ^2} the sum is of order X^2 (proved, Proposition 8.1), and for the μ^2-corner it is measured to be of order X^2. Measured: the part of the corner with $X/(dj) \le 64$, with $|\mu(d)\mu(j)|$ in place of $\mu(d)\mu(j)$, is $-0.00090X^2$, $-0.00121X^2$, $-0.00148X^2$ at $X = 30000.5$, 120000.5, 480000.5; the same part of $\mathcal{C}(X)$ itself is $+0.051X^{3/2}$, $+0.108X^{3/2}$, $-0.054X^{3/2}$ at these X. The first two agree with the sum of the bands $k \le 5$ of Table 2 (which lists $-\mathcal{C}$); $X = 480000.5$ is not in the tables. Writing W_χ for (5.1) with $\mu(n)$ replaced by $\mu(n)^2 \prod_{p|n} \chi(p)$, and fixing $\chi(p) = -1$ for $p \le \sqrt{X}$ and choosing the signs of the larger primes freely still allows $W_\chi(X)$ up to $0.055X^2$, $0.044X^2$, $0.042X^2$ at $X = 10^4$, 10^5, $2 \cdot 10^5$.

Parity. At $X = 1.2 \cdot 10^5$ let S_{even} and S_{odd} be the sums of $dj\, E_G(X/(dj))$, without the sign $\mu(d)\mu(j)$, over the squarefree pairs $d, j \le \sqrt{X}$ with $X/(dj) \le 64$ and $\omega(d) + \omega(j)$ even, respectively odd, where $\omega(n)$ is the number of prime factors of n. Then $S_{\text{even}} = -6.459 \cdot 10^6$ and $S_{\text{odd}} = -1.0958 \cdot 10^7$, with absolute sums $1.082 \cdot 10^9$ and $1.066 \cdot 10^9$, and the corresponding part of the corner is $S_{\text{even}} - S_{\text{odd}} = 4.499 \cdot 10^6 = 0.108X^{3/2}$.

Dyadic bands. Splitting $\mathcal{C}$ by $X/(dj) \in [2^k, 2^{k+1})$, single bands reach $0.73X^{3/2}$ in absolute value, larger than the total, and neighbouring bands $k = 1, 2$ have opposite signs in all three cases (Table 2).

13. Numerical observations

All computations are in double precision unless stated, with exact integer Möbius values; scripts and outputs are listed in Appendix A.

Table 1. The exact decomposition (6.10), in units of $X^{3/2}$; in parentheses the sum of the absolute values of the terms. The last column is the residual of (6.10).

X	W	$2W^\flat$	$-aX^2m^2$	$-\mathcal{R}$	$-\mathcal{C}$	residual
30000.5	−0.1579	−0.0004	−0.0006	−0.0245 (0.442)	−0.1323 (26.0)	$3 \cdot 10^{-8}$
120000.5	−0.0948	−0.0000	−0.0112	−0.0054 (0.421)	−0.0783 (53.5)	$3 \cdot 10^{-7}$
380000.5	−0.1052	+0.0001	−0.0000	+0.0091 (0.359)	−0.1142 (95.5)	$1 \cdot 10^{-6}$

Table 2. $-\mathcal{C}$ split into bands $X/(dj) \in [2^k, 2^{k+1})$, in units of $X^{3/2}$; in parentheses the sum of absolute values.

X	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
30000.5	+0.037 (4.8)	+0.273 (7.0)	−0.434 (6.8)	−0.065 (3.3)	−0.005 (2.1)	+0.143 (1.1)
120000.5	+0.206 (10.3)	+0.309 (14.3)	−0.725 (13.8)	+0.250 (6.7)	−0.116 (4.3)	−0.032 (2.2)
380000.5	−0.095 (18.7)	−0.252 (25.6)	+0.547 (24.5)	−0.249 (11.8)	−0.103 (7.7)	+0.080 (3.8)

Size of E_G. Over 20 random Y in $[10^q, 2 \cdot 10^q]$, the root mean square of $E_G(Y)/Y^{3/4}$ is 0.948, 1.119, 0.676 for $q = 3, 4, 5$. Over 4000 values of Y drawn log-uniformly in $[10, 2 \cdot 10^5]$, $\max |E_G(Y)|/Y^{3/4} = 2.830$, and over the first 400 of them $\max |E_G(Y)|/Y = 1.025$. The identity of Lemma 6.2(iii) was checked by numerical differentiation at $Y = 37.3$, 512.77, 4096.41 (agreement to 10^{-5}).

TABLE 3. Residues $r_k = r_{\rho_k}$ of $\widehat{W}$ at $s = 1 + \rho_k$ for the first five zeros (mpmath, 30 digits).

k	γ_k	$\lvert A(\rho_k)\rvert$	$\lvert\zeta'(\rho_k)\rvert$	$\lvert r_k\rvert$
1	14.1347251417	0.147337	0.79316	0.0821128
2	21.0220396388	0.112785	1.13684	0.0295771
3	25.0108575801	0.100191	1.37172	0.0183161
4	30.4248761259	0.0875278	1.30394	0.0138456
5	32.9350615877	0.0828281	1.38212	0.011421

Size of W and the zeros. Over 60 random X in $[10^q, 2\cdot 10^q]$, $q = 3, 4, 5$: $\max|W|/X^{3/2} = 0.2335$, 0.2827, 0.2346; root mean square of $W/X^{3/2}$: 0.133, 0.127, 0.141. The correlation of $W(X)$ with $2\operatorname{Re}\sum_{k\le 100} r_k X^{1+\rho_k}$ is 0.99985, 0.99993, 0.99986, with regression slopes 0.9975, 0.9953, 0.9976. For H, over 150 random x in $[10^q, 2\cdot 10^q]$, $q = 3, \dots, 6$, the root mean square of $H/\sqrt{x}$ is 0.399, 0.417, 0.413, 0.475, and the correlation with $2\operatorname{Re}\sum_{k\le 150} A(\rho_k)x^{\rho_k}/\zeta'(\rho_k)$ is 0.937, 0.935, 0.930, 0.951.

The no-go constant. $W_{\mu^2}(X)/X^2 = 0.0977$, 0.0975, 0.0968 at $X = 1000.5$, 10000.5, 100000.5, against $(6/\pi^2)a = 0.09673$ from (8.1).

Identities. (4.5) was checked at $x = 2.37$, 17.61 (agreement to 10^{-8}; at $x = 1234.57$ and 98765.43 the truncation $d \le 4\cdot 10^6$ of the series dominates the difference); (4.4) to $2\cdot 10^{-11}$; (4.3) for $2 \le n < 400$, evaluating H at distance 10^{-9} to the right of n (difference at most $1.2\cdot 10^{-4}$); Proposition 5.1(i) at $X = 57.3$, 300.7 (agreement to 10^{-6}) and (ii) at one value of X to 20 digits; Proposition 7.3 at $X = 20000.5$, 100000.5 (residual $< 10^{-5}X^{3/2}$).

14. What the route reduces the Riemann hypothesis to

By Theorem 6.4, RH is equivalent to the following inequality, which is not proved: for every $\varepsilon > 0$ there is K_ε such that

$$\sum_{\substack{d,j\le\sqrt{X}\\ dj>\sqrt{X}}} \mu(d)\mu(j)\,dj\,E_G\Big(\frac{X}{dj}\Big) \ge -K_\varepsilon X^{3/2+\varepsilon} \qquad (X \ge 1), \tag{14.1}$$

where $E_G(Y) = \sum_{\ell\le Y} \ell\,\mathrm{Cl}_2(2\pi Y/\ell) - \pi(C - \log 2)Y^2$. Only μ on $[1, \sqrt{X}]$ and E_G on $[1, \sqrt{X})$ enter. Inequality (14.1) is one-sided, and the term $-aX^2 m(\sqrt{X})^2$ of (6.10) plays no role in it. This document does not prove it. By Remark 8.2, a proof of it must use more about μ than the multiplicativity relation on squarefree integers.

By Proposition 11.1, (14.1) reads $\sum_{\sqrt{X}<n\le X} \lambda(n) b_{\sqrt{X}}(n)\, nE_G(X/n) \ge -K_\varepsilon X^{3/2+\varepsilon}$, a one-sided bound for a Liouville sum with the explicit nonnegative weight $b_{\sqrt{X}}$, and by Proposition 11.2 it is X times a weighted Liouville sum of dilations of the Müntz transform of the Clausen kernel. A second unproved statement, (9.12), is sufficient for RH but not known to be equivalent to it. It is a bound for a positive semidefinite quadratic form in $(\mu(d))_{d\le N}$ whose matrix (9.8) is explicit, depends on d, e only through $(de)^{-\sigma}$ and $\log(e/d)$, and has a diagonal bounded uniformly in N.

References

[1] L. Báez-Duarte, A general strong Nyman–Beurling criterion for the Riemann hypothesis, arXiv:math/0505453, version 1 (2005), `https://arxiv.org/abs/math/0505453` (only its abstract is used here). (Cited on p. 15.)

[2] H. Davenport, *Multiplicative Number Theory*, second edition, revised by H. L. Montgomery, Graduate Texts in Mathematics 74, Springer, New York, 1980, doi:10.1007/978-1-4757-5927-3. (Cited on p. 11.)

[3] W. Rudin, *Real and Complex Analysis*, third edition, McGraw-Hill, New York, 1987, ISBN 0-07-054234-1 (no DOI). (Cited on pp. 8, 11, 14, 15, 16, 17, and 18.)

[4] E. C. Titchmarsh, *The Theory of Functions*, second edition, Oxford University Press, Oxford, 1939 (no DOI). (Cited on p. 17.)

[5] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, second edition, revised by D. R. Heath-Brown, Clarendon Press, Oxford, 1986 (no DOI). (Cited on pp. 4, 5, 8, 10, 11, 12, 14, 16, and 18.)

Appendix A. Code and data

The zip file that accompanies this document contains every script with its output (`X.py` and `X_output.txt`, and a copy `X.py.txt`). The constants of Sections 2–5 and Table 3 come from `route_constants.py`. Table 1 and the size of E_G come from `corner_exact.py` and `bilinear_W.py`; Table 2 from `corner_bands.py`; parity, $E'_G = -2\pi R$ and the restricted corner from `corner_parity.py` and `corner_scale.py`; W against the zeros from `mu_clausen.py`; the no-go constant and the χ experiments from `nogo_chi.py`; Proposition 7.2 from `lcm_form.py` and `redheffer_W.py`; Proposition 7.3 from `huxley_watt3.py`; (5.2) from `W_kernel2.py`; Proposition 4.2 and (4.3) from `telescoping.py`, `idcheck.py` and `left_sing.py`; the explicit formula for H from `H_explicit.py`; Section 11 from `liouville_corner.py`, `liouville_weight.py`, `corner_onesided.py` and `musq_corner_const.py`. The remaining scripts of the folder `pairs` belong to earlier exploration of the same functions and are included unchanged.

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Statements reviewed by Claude: 37.